



Matthias Templ

Curriculum Vitae

Personal Data

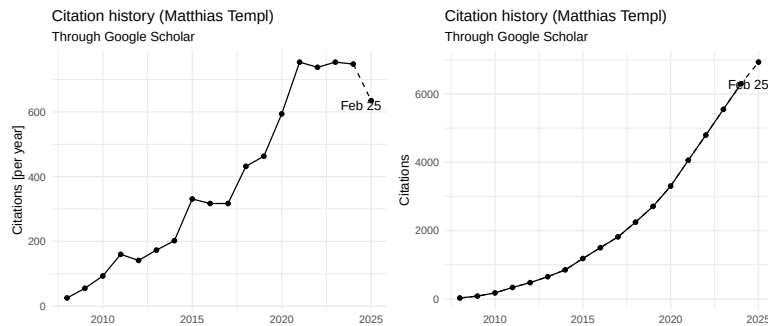
Nationality Austria

Education and qualification

- 2022 **Professor in Empirical Methods and Business Analytics**, *University of Applied Sciences and Art, Northwestern Switzerland.*
- 2020 **Qualification in university didactics**, *Zurich University of Applied Sciences.*
- 2013 **Venia docendi (habilitation) in Statistics**, *Vienna University of Technology.*
- 2009 **PhD in Technical Mathematics**, *Vienna University of Technology*, (with excellence and awarded).
- 2003 **Master in Technical Mathematics / Mathematics in Economics**, *Vienna University of Technology*, (with excellence).
- 1998 **General qualification for university entrance**, *Johannes Kepler University Linz.*

Publication statistics

- GS **Citations:** 7 650 (since 2021: 4 169). **H-index:** 38 (since 2021: 29). **i10-index:** 84 (since 2021: 55). (accessed June 2026)
- Researchgate Research Interest Score 2982, **Citations:** 5006, **H-index:** 33 (figures predate the Google Scholar update above)
- ORCID 0000-0002-8638-5276



Current and former positions

- 2022-ongoing **Professor in Statistics**, *Institute for Competitiveness and Communication, University of Applied Sciences, Northwestern Switzerland, Switzerland.*
- 2021-ongoing **Lecturer**, *Plant Science Center, ETH Zurich, Switzerland.*
- 2020-2022 **Fellowship**, *Digital Initiative of the Canton of Zurich. Anonymization, Switzerland.*
- 2016-2022 **Lecturer and researcher**, *Institute of Data Analysis and Process Design, Zurich University of Applied Sciences, Switzerland.*
- 2016-ongoing **Assigned through habilitation; Supervisor, Lecturer**, *Institute of Statistics & Mathematical Methods in Economics, Vienna University of Technology, Austria.*
- 2013-2016 **Associate Professor**, *Institute of Statistics & Mathematical Methods in Economics, Vienna University of Technology, Austria.*
- 2011-2022 **Owner and founder of the company**, *data-analysis OG, Austria.*
- 2011-2016 **Consultant**, *UNIDO and World Bank.*
- 2004-2016 **Research Scientist**, *Quality and Methods Unit, Statistics Austria, (1/2 position).*
- 2012-2015 **Consultant**, *Palacký University Olomouc, (1/2 position).*
- 2009-2012 **University Assistant, Post-Doc.**, *Department of Statistics and Probability Theory, Vienna University of Technology (1/2 position).*
- 2004-2009 **University Assistant**, *Department of Statistics and Probability Theory, Vienna University of Technology, 2/3 position.*
- 2002-2003 **Research Scientist**, *voestalpine, R & D department, Linz, Austria.*

Research visits

- regularly **Palacký University of Olomouc**, *Czech Republic*, Collaborator: Karel Hron.
- regularly **Vienna University of Technology**, *Austria*, Collaborator: Peter Filzmoser.

- Aug 2019 **Gadjah-Mada-University (UGM)**, *Indonesia*, Collaborator: Dedi Rosadi.
- Sept 2016 **University of Albacete**, *Spain*, Collaborator: Virgilio Gómez Rubio.
- Dec 2014 **Technical University of Graz**, *Austria*, Collaborator: Herwig Friedl.
- Mai 2013 **World Bank**, *Washington, USA*, Collaborator, Olivier Dupriez.
- June 2012 **OECD and Paris21**, *Paris, France*, Collaborator: Francois Fontenau.
- June 2011 **University of Trier**, *Germany*, Collaborator: Ralf Münnich.
- June 2010 **Applied University of Northern-Switzerland**, *Olten, Switzerland*, Collaborator: Beat Hulliger.

Membership of associations, editorships and awards

- Editor-in-Chief Austrian Journal of Statistics, <http://www.ajs.or.at>, great performance since the editorship (2014) ([Scopus data-base statistics](#))
- Assoc. Editor Journal Wirtschafts- und Sozialstatistische Archiv (from Springer)
- Sec. Editor Algorithms
- Adv. Board Services for research data at the Zurich University of Applied Sciences (2020 - 2022)
- Elect. Memb. International Statistical Institute (ISI) (2024 - ongoing)
- Pres. Board Austrian Statistical Society (2014 - ongoing)
- Board Memb. Swiss Statistical Society, section Official Statistics
- Jury Selection committee for the Lambert Award from the Swiss Statistical Society
- Maintainer CRAN Task View on Official Statistics & Survey Statistics
- Maintainer CRAN Task View on Compositional Data Analysis
- Mentor Google Summer Schools of Code in 2012, 2013, 2014, 2016
- Award Austrian Society for Statistics for the best PhD thesis in applied/computational statistics 2009
- Award Winner of the World First Class Professorship programme from the Indonesian Government in 2019.
- Award Springer BestMasters (Verena Puchner), 2014. Supervisor.
- Award Austrian Society for Statistics for the best master thesis in applied/computational statistics in (Marlene Weinauer), 2017. Supervisor.
- Award ISI Young Statisticians Paper Award (Marlene Weinauer), 2019. Supervisor.
- Award Winners of the 2020 Young Statisticians' Prize. Supervisor of my PhD students work, Johannes Gussenbauer and Mr. Gregor de Cillia.

Organisation of scientific events

- **Organizer** of the
 - *Survey Conference (SurvConf25)* in Olten Switzerland, 2025.
 - *Workshop Population Health Data Anonymisation* at MEIRU, Malawi, 2021.
 - *Workshop zur Anonymisierung von personenbezogenen Einzeldaten* (Workshop on data anonymisation) in Biel, Switzerland, 2017, 2018 and 2019.
 - *Survey Conference (SurvConf17)* in Gumpoldskirchen, Austria, 2017.
 - *Summer School on Advanced R* in Vorau, Austria, 2016
 - *IASC seminar on R* held in Dar Es Salaam, Tanzania, 2015. (together with Peter Filzmoser)

- *Summer School on Advanced R* in Vorau, Austria, 2013.
- *International Workshop on Statistical Disclosure Control* at OECD in Paris, 2012.
- *International Conference on Indicators and Survey Methodology*, February 24.-26., 2010, held at the Vienna University of Technology.
- *International Workshop on Exploratory Data Analysis and Visualization (EDAVIS 2010)* in May 27.-28, 2010, Vienna.
- o **Local-Organizer** of the
 - *Annual and 50th anniversary conference of the International Association for Mathematical Geosciences (IAMG 2018)* in Olomouc, Czech Republic
 - *Statistische Woche 2012* in Vienna (530 participants).
- o **Local-Organizer and Host** of the *ViennaR Meetup* (on monthly basis from 2015-2017).
- o **Co-Organizer** of the
 - *19th Annual Conference of the International Association for Mathematical Geosciences (IAMG 2018)* in Olomouc, 20 year anniversary, 2018.
 - *Swiss Days of Statistics* in Zurich, 125 year anniversary, 2018.
 - *European Statistical Training Programme Course on R in Official Statistics* in Vienna, 2016.
 - *GeoMap 2014 International Workshop* in Olomouc.
 - *International Conference CoDaWork 2013 on Compositional Data Analysis (CoDaWork 2013)* in May 2013 in Vorau.
- o **Member** of the **scientific committee** at the (incomplete list):
 - *International Conference on Computing, Mathematics, and Statistics (iCMS)*, Kedah, Malaysia (2025).
 - *SurvConf 2025*, Olten, Switzerland (2025).
 - *Statistische Woche*, Vienna (2024).
 - *International Conference on R in Official Statistics (uRoS)* in Bucharest, Romania (2015, 2016, 2017, 2018, 2022, 2024), Voorburg, Netherlands (2019), Vienna (2021).
 - *Computational Data and Modelling* in Minsk, Belarus (2019, 2022).
 - *Data Privacy Management (DPM 2016, 2017, 2018)*.
 - *The Statistical Week (Statistische Woche Österreich 2018)* in Linz, Austria, 2018.
 - *The Swiss Statistics Days (Statistische Woche Schweiz 2018)* in Zürich, Switzerland, 2018.
 - *The 7th International Workshop on Compositional Data Analysis (CoDaWork 2017)* in San Salvatore, Italy, 2017.
 - *New Techniques and Technologies in Statistics (NTTS)* conference in Brussels, 2017.
 - *The 9th WISTP International Conference on Information Security Theory and Practice (WISTP'2015)*, Heraklion, Greece, 2016.
 - *Computational Data Analysis and Modelling (CDAM)* in Minsk, Belarus, 2016.
 - *13th IEEE International Conference on Advanced and Trusted Computing (ATC 2016)* in Toulouse, France, 2016.
 - *CoDaWork International Workshop* in Girona, Spain, 2015 and 2018 for the selection process of CoDa conferences. *useR!2013* conference in La Mancha, Spain, 2013.
 - *UNECE worksession on data editing* in Ljubljana, Slovenia, 2011 and Oslo, Norway, 2012.

Keynote presentations, courses and workshops

- o **Session chair** at many conferences, e.g. CoDaWork in Girona 2024, the The European R User Meeting (eRum) in Poznan 2016, International Conference on Official Statistics and R in Bucharest/Romania 2014, 2015 and 2016, missData conference in Rennes/France 2015, International Workshop on Survey Methodology and Official Statistics (SurvConf) in Berlin 2015, International Workshop on Practical Aspects of Geochemical Exploration and Mapping with Logratio Techniques (GeoMap) in Olomouc/Czech Republic 2014, International Conference of the ERCIM WG on Computational and Methodological Statistics in London/UK 2014, useR2013! in Los Angeles/USA, International Workshop on Compositional Data analysis (CoDaWork) in

Spain 2011 and 2015 and Vorau/Austria 2013, UNECE worksession on data editing 2011 in Lubjiana, 2012 in Rome and 2013 in Oslo, the International Conference on Indicators and Survey Methodology 2011 and EDAVIS 2011 in Austria.

- **Keynote speaker at conferences and larger scientific events:** Anonymization and Synthetization of Data, Swiss Clinical Trial Annual Meeting, Bern, Switzerland (2024); Juristinnentag (2022), Zurich; Informatik, 51. Jahrestagung der Gesellschaft für Informatik (2021); Datenschutzrechtstagung (2021), Zurich; Use of R in Official Statistics (uros 2020) in Vienna; International Conference on Computing, Mathematics and Statistics (iCMS2017, iCMS2019, and iCMS2021) in Langkawi Island, Malaysia; Symposium on Anonymisation of the Association for Data Protection Switzerland (2019); European R User Meeting (eRum 2016) in Poznan, Poland; the International Conference on Official Statistics with R (2015) in Bucharest, Rumania; the *f.eh im Dialog* (Forum Ernährung Heute) (2015) in Vienna, Austria; Conference on Teorie a praxe statistického zpracování dat (2014, 2015) in Czech Republic; Hackers Hack Meetup (2014) in Vienna, Austria; the International Conference on Indicators and Survey Methodology (2010) in Vienna, Austria; the Data Validation Day (2009) in Luxembourg.
- **Given Tutorials/Courses/Workshops**, e.g. *Working with sdcMicro: Anonymizing a Real Dataset* at the Data Protection Day, University of Zurich in 2026; *Anonymization and Pseudonymization of Research Data* (2-day course for data stewards and researchers) for the Higher Education and Science Information Technology Sharing Service Centre (VPC), Latvia, in 2026; *Anonymization and Synthetic Data – Methods, Tools, and Applications* in Olten, Switzerland in 2026; *Anonymization with Statistical Disclosure Control Methods using sdcMicro* at the Data Protection Week, University of Zurich in 2025; *Disclosure Risk of Anonymized and Synthetic Data* at the Clinical Trial Center Zurich in 2024; *Anonymization of data with sdcMicro* at the Data Protection Week, University of Zurich in 2024; *Visualization and Imputation of Missing Value* at DaSS, University of Lisboa in 2024, *Anonymization of data with sdcMicro* for EU CLARIN at University of Basel in 2024, *Introduction to data anonymization* at University of Basel in 2024; *On-the-job training on Anonymization of Population data* for World Bank in 2022 in Bucharest; *Anonymization of Census data* for World Bank in 2021 (online); *Anonymization of personal data with applications in R* at the Zurich R courses from the University of Zurich; *Data Anonymization Workshop* for the Inter-American Development Bank in 2021; *Population Health Data Anonymisation Workshop* at MEIRU in 2021; on *Anonymization* at SBB (2 workshops, 2020); *Data Visualization in R* at the Summer School on R and Data Science at the University of Bolzano 2019; *Data Visualization of tidy data* at the iCMS 2019 in Langkawi, Malaysia; *Data Visualization* at the University of Gadjah-Mada in Yogyakarta, Indonesia in 2019; *R* at the Austrian Institute of Economic Research 2018, on *R and data manipulation: big data applications* in Malaysia 2017; Statistical Disclosure Control (*ANON-Schul: Schulung zur Anonymisierung von personenbezogenen Datenbeständen*) in Biel 2017, 2018 und 2019; *Advanced Graphics in R* at the European Academy of Bozen/Bolzano 2017, *Small Area Estimation and Compositional Data Analysis* at the GEOSTAT Summer School in Albacete/Spain 2016; *Introduction to Data Visualization* at the European R Users Meeting in Poznań 2016; *The Use of R in Official Statistics* European Statistical Training Programme 2016; *Introduction to R* at REWE Austria 2016; *R course* for the National Bank Austria 2016; *R courses* for the Chamber of Commerce 2016, 2017, 2018; *Skill Gap Survey with R* in Abuja/Nigeria 2015; *Data Visualization* at the School on Data analysis and programming with R in Bozen/Italy 2015; *Introduction to Statistics with R* in Dar Es Salaam/Tanzania 2015; *Advanced features of R* at Böhringer-Ingelheim 2015; *R in the Statistical Office* at the Quality in Official Statistics Conference (Q2014); *Introduction to R* at the Austrian Chamber of Commerce 2012, 2013 and

2014; *Using R in Official Statistics* at the New Techniques and Technologies in Statistics (NTTS 2015) conference in Brussels/Belgium; *Statistical Disclosure Control* at OECD in Paris 2013; *R in the Statistical Office* at Statistische Woche 2012.

Projects

2026	Anonymization Process Development for Société des Produits Nestlé SA (Service Agreement RDLS102071/CMT5146)	16.5k CHF
2025 - 2027	Privacy-Preserving Data Sharing for the SBB Synthetic Population — Anonymization, Geographic Protection, and Disclosure Risk Assessment (Oct start, ~1.5 years)	60k CHF
2024 - 2028	SNF Project: Epidemic source detection on contact networks - from theory to practice (co-applicant)	392k CHF
2023 - 2027	SNF Bridge Discovery Project: Harnessing event and longitudinal data in industry and health sector through privacy preserving technologies	710k CHF
2022	Anonymisation of Census data for World Bank	15k CHF
2021	Training Statistical Disclosure Control for World Bank	19k CHF
2021	Anonymization of data for the Inter-American Development Bank	6k CHF
2021	EnerSuisse grant on anonymization of smart-meter data	12k CHF
2019 - 2021	Several contracts with the Free University of Berlin and University of Bamberg for the creation of online materials for courses, especially online tests and tasks in R; creation of R tutorials; co-conception of e-exams based on R.	8k CHF
2020	ZHAW Internal Grant on imputation with deep learning methods (follow-up grant)	16k CHF
2020	Grant on a Fellowship <i>Digital Initiative from Kanton Zurich on Anonymization</i>	90k CHF
2020	Start-Up Grant from ESTHER Switzerland <i>Better de-Identification using Statistical Anonymisation (BISA) for Malawi population health data</i>	10k CHF
2020	Grant from SBB <i>Anonymisation of SBB movement data</i>	18k CHF
2020	Grant from Swisscom <i>Anonymisation of trajectory data</i>	18k CHF
2019	Grant from EMC Research <i>R package for table generation</i>	€ 15k
2019	Strategic Grant from IDP/ZHAW. Imputation using deep learning methods	11k CHF
2019	Grant from IDP/ZHAW. Imputation with artificial neural networks	16k CHF
2019	Grant from the Indonesian Government <i>World First Class Professorship Programme</i>	12k CHF
2019	Grant from Helsana Gesundheitswissenschaften <i>Anonymisation of data from Helsana</i>	37k CHF
2018	Grant from SIA (Chamber of Engineers and Architects Switzerland). <i>Statistical Modelling of remunerations</i>	38k CHF
2018	InnoSuisse project about bee hive data analysis (granted but not launched)	220k CHF
2018	IDP/ZHAW grant on imputation	24k CHF
2017	ZHAW grant on digital learning project <i>Interaktives, automatisches Feedbacksystem für den Praktikumsunterricht in der Datenanalyse</i>	37k CHF
2016	World Bank grant, selection number 7179283. <i>Re-designing the GUI for sdcMicro</i>	€ 35k
2016	Special grant agreement for statistical disclosure control, Eurostat. <i>"Tools for Statistical Disclosure Control"</i>	€ 25k
2016	World Bank grant, selection number 1206769. <i>Synthetic Data for Microsimulation</i>	€ 33k
2015	World Bank grant, selection number 1129231. <i>"Methods and tools for the generation of synthetic populations"</i>	€ 31k

2015	World Bank Fund for Statistical Capacity Building. <i>"IASC seminar on R held in Tanzania"</i>	€ 15k
2015	UNIDO. <i>"The Skill Gap Survey in Nigeria"</i>	€ 27k
2014 - 2015	World Bank grant, selection number 1157976. <i>"Improving the quality of sample household expenditure data and the reliability of poverty and inequality measures"</i>	€ 38k
2014 - 2015	Special grant agreement for statistical disclosure control, Eurostat. <i>"Generating Public-Use Files for EU-SILC and LFS"</i>	€ 43k
2014	Analysis of data bases and analysis of incidence from Wiener Linien	€ 17k
2008 - 2013	Various of projects of about €60k for the OECD and the International Household Survey Network on statistical disclosure control	€ 60k
2011 - 2012	Subcontract for the European project <i>ESSnet on common tools and harmonised methodology for SDC in the ESS</i> , Eurostat grant	€ 20k
2010	European project <i>Exploratory Data Analysis and Visualization (EDAVIS)</i> , Eurostat grant	€ 48k
2008 - 2011	Advanced Methodology for European Laeken Indicators (AMELI), project number 217322, coordinator: Ralf Münnich, Uni Trier. Funding from the European Commission under the 7-th framework programme for research	€ 265k
2008 - 2010	European project <i>"ESSNet on Statistical Disclosure Control"</i> , grant agreement no 25200.2005.003-2007.670. Funding from the European Commission/EUROSTAT	€ 22k
2006	European project <i>"Centre of Excellence in Statistical Disclosure Control"</i> , grant agreement no 25200.2005.001-2005.619. Funding from the European Commission/EUROSTAT	€ 22k
2009 - 2018	3 projects with UNIDO (€4k each), the Austrian Federal Ministry of Agric., Forestry, Env. and Water Manag. (biodiversity) (€3k), Pollen information of Austria (demographic analysis of symptoms) (> €15k), mobilkom (automatic reporting) (€13k), the Austrian antitrust authority (low-cost carriers analysis) (€2.5k), complicity in medical studies (> €4.5k), traffic modelling (€12k), 3× Google Summer School of Code, various participations in European projects, Aut. and interactive analysis of questionnaires for ooselo GmbH, Switzerland (€3700), CoDa book (€9.4k), Entwicklung eines stationsungebundenen E-Bike-Sharing- Services (contribution ≈ €68k from €910k), IDP/ZHAW grant on compositional data analysis (€9k)	<€ 15k

Software development in R

Maintainer of **R**'s CRAN Task View on *"Official Statistics and Survey Methodology"*, <http://cran.r-project.org/web/views/OfficialStatistics.html>

Developed R packages:

(the code on bitbucket is only available upon request)

- clustTool** Graphical user interface for clustering data. Several new visualisation methods for clusters. On CRAN (archived)
- laeken** Estimation of indicators on social exclusion and poverty. On CRAN. Published in the Journal of Statistical Software (JSS).
- mobi** Tools for simulation and modelling free-floating vehicle sharing systems. On bitbucket

robCompositions	Robust multivariate analysis of compositional data. On CRAN and github. Main software for the book Applied Compositional Data Analysis. With Worked Examples in R (Springer Series in Statistics).
sdcmicro	Software for microdata protection to provide anonymised data to researchers. On CRAN and github. Published in JSS. State-of-the-art software in data anonymization.
simPop	Simulation of synthetic data. On CRAN and bitbucket. Published in Journal of Statistical Software.
sparkTable	Sparklines and graphical tables for documents and websites. On CRAN (archived) and github. Published in the R journal.
TGUIColor	Teaching GUI and modern blended learning feedback system for the generation of interactive lectures. Core functionality. on CRAN (archived). Published in the Journal of Statistical Software.
TGUITeaching	Teaching GUI and modern blended learning feedback system for the generation of interactive lectures. Prototype. On CRAN (archived)
VIM	Visualization and imputation of missing values. On CRAN and github. Published in the Journal of Statistical Software. More than 1.7 million downloads (through RStudio server only).

Under my supervision or contributed:

emdi	Small area estimation and mapping regional/spatial disaggregated indicators. On CRAN. Published in the Journal of Statistical Software.
simFrame	A general framework for statistical simulation. On CRAN. Published in the Journal of Statistical Software.
tguishiny	A shiny implementation of an interactive, automatized blended learning and feedback tool. On bitbucket
X12	Wrapper functions, interactive timeline management of parameters and results and a GUI for the X12 binaries. Archived. Published in the Journal of Statistical Software.

--- Software and other Skills (selection)

Basic	MATLAB, MAPLE, MATHEMATICA
Intermediate	TCL/TK, GTK2, HTML/PHP, SAS, SPSS, C, C++, SED, UNIX, GIT
Advanced	Advanced R development, L ^A T _E X, MARKDOWN, R SHINY
Advanced	Leadership skills. By leading dozens of projects, but also from certificates, such as from Julian Birkinshaw “Managing the Company of the Future” (London School of Economics)
Advanced	Didactics skills: Evaluation and certificate by ZHAW and venia docendi from TU Vienna.

--- Teaching

On regular basis at universities:

Statistical Modelling	Classical and robust linear methods, variable selection, GAM (in English)	ETH Zurich, PhD programme Plant Sciences
Compositional Data Analysis	Introduction to compositional data analysis for ecologists (in English)	ETH Zurich (first time in Jan. 2022)
Seminar on computational statistics	Changing subjects on computational statistics (in English)	Vienna Univ. of Techn.
Diploma and dissertation seminar	Changing subjects in statistics (in English)	Vienna Univ. of Techn.

On regular basis at applied universities:

Business Statistics	Introduction to statistics (in English)	University of Applied Sciences, Northwestern Switzerland
Statistics and probability theory	Introduction to statistics	University of Applied Sciences, Northwestern Switzerland
Statistics for International Management	Introduction to statistics (in English)	University of Applied Sciences, Northwestern Switzerland
Managerial Data Science	Multivariate Statistics and artificial intelligence	University of Applied Sciences, N-W. Switzerland

Former teachings at universities:

Advanced Survey Statistics: Statistical Disclosure Control	Disclosure risk of confidential data, anonymization, data utility, spatial and trajectory data, privacy preserving comp., federated learning (in English)	Free Univ. of Berlin, Univ. Trier and Univ. of Bamberg
Regular seminar talks for PhD students and staff	from 2012 - 2015. Approx. every two months	Palacký University of Olomouc
Statistical Computing	Introduction to statistical analysis using modern computer methods with an emphasis on applications and software R. Turning new ideas and methods into software. Monte-Carlo methods.	Techn. Univ. of Vienna
Statistics for Aviation	Introduction to statistics including statistical inference, testing and regression analysis.	ZHAW

Cluster Analysis and Methods for Unsupervised Learning	Theoretical concepts and application of cluster analysis and methods for unsupervised learning. Practical applications using the software environment R.	Techn. Univ. of Vienna
Legal Aspects of Statistical Methods	Realization of legal consequences when applying statistical methods. Applications in Official Statistics and Biostatistics.	Techn. Univ. of Vienna
Official Statistics and Sampling Techniques	Statistical methods in official statistics. Editing, Imputation, Complex Sampling Designs, Small Area Statistics, Statistical Disclosure Control, Statistical Matching,	Techn. Univ. of Vienna
Computational Statistics	Analysis and statistical inference. Analysis of variance and regression analysis using modern statistical methods.	Techn. Univ. of Vienna
Stat. Simulation and Computer-Intensive Methods	Generation of random numbers. Statistical simulation. High-performance computing and resampling methods. MCMC. System dynamics. Optimization. Micro-simulation.	Techn. Univ. of Vienna
Research Methods in Computer Science	Survey methodology and formal methods of research in computer science. Literate programming, automatization, design of experiments and surveys, applications in medical science, official statistics, geoscience, etc.	Techn. Univ. of Vienna
Fundamental research methods for doctoral students	Quantitative methods for computer scientists, survey design, basic statistics, resampling.	Techn. Univ. of Vienna
Seminars, Practicals, Courses	Related to topics Statistics and Probability Theory, Computational Statistics and Official Statistics.	Techn. Univ. of Vienna

Former teachings at applied universities:

CAS Data Analysis	Module PCA, Cluster Analysis and Classification/Machine Learning	Zurich University of Applied Sciences
Survey Design and Analysis	Survey design, survey statistics, analysis of complex surveys	Zurich University of Applied Sciences (ZHAW)
Statistical Modelling	Basic and advanced methods in statistical modelling (in English)	ZHAW
Statistics for Engineers 3	Statistical inference and testing	ZHAW

Continuing education studies (CAS), Data Analysis	Module Descriptive statistics, visualization and R	ZHAW
CAS Big Data, Blockchain and Distr. Ledger	Data analysis and visualisation	ZHAW

Former teachings at Statistics Austria:

(lectures in statistics are supported with the self-developed blended learning TGUI system)

Introduction to Statistics and Probability Theory	Descriptive Statistics. Probability Theory. Data Visualization.
Survey Methodology and Official Statistics	Survey Design, Complex Estimation with Surveys, Multi-stage and Cluster Designs, Calibration and Non-Response Calibration, Bootstrap and Calibrated Bootstrap in Survey Statistics.
Multiple and Multivariate Methods in Official Statistics	Regression. Editing and Imputation. Time Series Analysis. Seasonal Adjustment. Record Linkage and Statistical Matching.
Statistical Disclosure Control	Microdata and Macrodata protection. Risk and utility. Synthetic data simulation.
Introduction to R	Basic R. Data Types. Data Manipulation. Basic Graphics.
Advanced Data Manipulation with R	Efficient data manipulation with the packages dplyr and data.table.
High Performance Computing	Parallel Computing. Profiling. Debugging. C++ integration
Data Visualization	Graphics system of R. Packages ggplot2, ggmap, tmap, sparkTable, ggvis.
Survey Analysis with R	Stratified, multistage, cluster designs in R. Estimation of complex indicators.

External courses:

Courses for companies	Courses and tutorials about R and development in R , visualization, survey methodology for the chamber of economics, Austrian Institute of Economic Research, Böhringer-Ingelheim, national bank of Austria, REWE, SBB, and various other enterprises.
Tutorials, Summer Schools and Workshops	On survey methodology, visualization, R, statistical disclosure control / anonymization, see above (under Workshops)

Commission work and supervision of students

- Commission member for the habilitation (postdoctoral lecture qualification) of PhD MMag.rer.soc.oec Klaus Prettner (Mathematical Economics), Vienna University of Technology, 2014.
- Commission member and reviewer for PhD defences several candidates in Czech Republic and Austria.
- Selection committee member for the position *Professor in Applied Statistics* at the Vienna University of Technology, 2015.
- Reviewer for the position *Professor für Statistik* at the Ludwig-Maximilians-Universität München (LMU), 2019.
- Reviewer for several grants and positions, e.g., for the *Strategy for Excellence Grant of the University of Tübingen* and National Science Centre Poland.
- A series of Bachelor thesis, master thesis and PhD thesis have been finished under my supervision. Currently, supervision of three PhD students.

Review work

Reviewer for the following 65 peer-review journals and book publishers (in alphabetic order, incomplete list):

Acta UP Mathematica, Advances in Statistical Analysis, Algorithms, Applied Geochemistry, AStA - Sozialwissenschaftliches Archiv, Austrian Journal of Statistics, BMC Medical Research Methodology, Canadian Journal of Statistics, Communications in Statistics, Communications in Statistics - Simulation and Computation, Computational Statistics and Data Analysis, Computers & Geosciences, Computer Graphics Forum, Data Science Journal, Economics Bulletin, Frontiers in Applied Mathematics and Statistics, IEEE Transactions on Knowledge and Data Engineering, Information, Information Visualization, International Journal of Data Science and Analytics, International Journal of Information Technology & Decision Making, International Journal of Management Science and Engineering Management, International Journal of Pattern Recognition, IT professionals, Journal of Applied Statistics, Journal of Computational and Graphical Statistics, Journal of Environmental Informatics, Journal of Hydrologic Engineering, Journal of Informations Sciences, Journal of Machine Learning Research, Journal of Official Statistics, Journal of Operations Research, Journal of Personalized Medicine, Journal of Probability and Statistics, Journal of Statistical Software, Journal of Statistical Computation and Simulation, Journal of Survey Statistics and Methodology, Journal of Systems and Information, Journal of Systems and Software, Journal of Water Resources and Protection, Journal of Water Research, Mathematical Geosciences, Mathematics, Metrologia, Mine Water and the Environment, Multivariate Behavioral Research, National Science Center Poland, Natural Resources Research, Oldenburg Verlag, Qeios, Ore Geology Reviews, PeerJ Computer Science, Remote Sensors, RJournal, Romanian Statistical Review, Sage, Scientific Reports, Springer Festschrift for Vera Pawlowski-Glahn, Software: Practice and Experience, Statistics, Statistical Journal of the IAOS, Statistical Papers, Statistics & Probability Letters, Survey Methodology, Transactions on Data Privacy, Transactions on Knowledge and Data Engineering, Transactions on Knowledge Discovery from Data, Wiley books.

Reviewer for the Polish National Science Centre, the National Science Center in Luxembourg, Eurostat and several conference proceedings (uRos, ISSTEC, ...). Evaluator for the Dutch Research

Council (NWO), Open Science Infrastructure round 2024–2025.

Reviews for proceedings papers at conferences are not listed, but could be requested.

Collaborators

Antonella Bucciante (University of Florence), Olivier Dupriez (World Bank), Jose Antonio Martin-Fernandez (University of Girona), Juan José Egozcue (Polytechnic University of Catalonia, Barcelona), Peter Filzmoser (Vienna University of Technology), Francois Fontenau (OECD), Virgilio Gómez Rubio (Universidad de Castilla-La Mancha), Karel Hron (Palacký University Olomouc), Beat Hulliger (University of Applied Sciences, North-Western Switzerland), Alexander Kowarik (Head of Methods Unit, Statistics Austria), Thomas Kuenzer (MedUni Graz), Kamarul Mansor (UiTM Malaysia, Kedah), Bernhard Meindl (Statistics Austria), Ralf Münnich (Universität Trier), Javier Palarea-Albaladejo (University of Girona), Vera Pawlowsky-Glahn (President International Association for Mathematical Geosciences (IAMG), University of Girona), Andreas Quatember (Johannes Kepler University Linz), Andreas Ruckstuhl (ZHAW), Murat Sariyar (FH Biel), Timo Schmid (Freie Universität Berlin), Tibor Standovar (ELTE), Carolin Strobl (University of Zurich), Barbara Templ (ETH Zürich), Valentin Todorov (formerly UNIDO), Nicolas Tzavidis (University of Southampton), Matthew Welch (World Bank), Vivian Zufferey (Agroscope)

References

Books

- [1] Matthias Templ. *Visualization and Imputation of Missing Values*. Springer Publishing, Cham, Switzerland, 2023.
- [2] Matthias Templ. *Statistical Disclosure Control for Microdata - Methods and Applications in R*. Springer International Publishing Switzerland, Springer International Publishing, 2016.
- [3] Matthias Templ. *Simulation for Data Science with R*. Packt Publishing, London, 2016.
- [4] Matthias Templ. *New Developments in Statistical Disclosure Control and Imputation: Robust Statistics Applied to Official Statistics*. Südwestdeutscher Verlag für Hochschulschriften, 2009.
- [5] Andreas Quatember and Matthias Templ. *Datenqualität in Stichprobenerhebungen: Eine verständnisorientierte Einführung in die Survey-Statistik*. Statistik und ihre Anwendungen. Springer Spektrum, Berlin, Heidelberg, 4 edition, 2026. in press.
- [6] Peter Filzmoser, Karel Hron, and Matthias Templ. *Applied Compositional Data Analysis. With Worked Examples in R*. Springer Series in Statistics, Cham, Switzerland, 2018.

Editor of Special Issues

- [7] Matthias Templ, editor. *Special Issue Statistical Disclosure Control for Microdata*. Special Issue of Algorithms, MDPI, Basel, 2019.
- [8] Karel Hron, Peter Filzmoser, and Matthias Templ, editors. *Proceedings of the 5th International Workshop on compositional Data Analysis CoDaWork 2013 June 3-7, 2013, Vorau, Austria*. TU WIEN, 2013.

- [9] Peter Filzmoser and Matthias Templ. *Special Issue - 10th International Conference Computer Data Analysis & Modelling 2013, Minsk, Belarus (Vol.43, 3-4)*. Österreichische Zeitschrift für Statistik. Österreichische Statistische Gesellschaft, 2014.
- [10] Kamarul Ariffin Mansor, Ciprian Alexandru, and Matthias Templ. Austrian Journal of Statistics. Vol 49 (3), Special Issue iCMS 2019. *Austrian Journal of Statistics*, 49(3), 2020.
- [11] Matthias Templ. Editorial special issue on official statistics. *Austrian Journal of Statistics*, 43(1):1, 2014.

Articles

- [12] Matthias Templ, Oscar Thees, and Roman Müller. RAPID: Risk of attribute prediction-induced disclosure in synthetic microdata. Technical Report arXiv:2602.09235, arXiv preprint, 2026. preprint.
- [13] Matthias Templ and Michael Stadler. Factors for analytical and intuitive cognition in strategy consultants: A multivariate analysis. *Acta Psychologica*, 263, 2026.
- [14] Matthias Templ. Log-ratio propagation on the simplex: A theory of cellwise contamination for compositional data. Technical Report arXiv:2605.31345, arXiv preprint, 2026. preprint.
- [15] Oscar Thees, Roman Müller, and Matthias Templ. Beyond the trade-off curve: Multivariate and advanced risk-utility maps for evaluating anonymized and synthetic data. Technical Report arXiv:2510.23500, arXiv preprint, 2025. preprint.
- [16] Matthias Templ and Christian Hofer. Compositional analysis of the relationships between organic matter content and chemical/physical properties of soil. *Applied Geochemistry*, 2025.
- [17] Barbara Templ, Matthias Templ, Anouk N' Guyen van Chinh, and Urs Brändle. Leveraging sankey diagrams for enhanced curriculum planning in higher education. *The Curriculum Journal*, 36(1):202–206, 2025.
- [18] Petra Jašková, Javier Palarea-Albaladejo, Karel Hron, David Lachman, and Matthias Templ. OPALS: Orthonormal pairwise logratio selection for compositional data in high dimensions. *Bioinformatics Advances*, 5(1), 2025.
- [19] Matthias Templ. Robust multiple imputation with GAM. *Statistics and Computing*, 34(4):121, 2024.
- [20] M. Templ and Markus Ulmer. The impact of misclassifications and outliers on imputation methods. *Journal of Applied Statistics*, 0(0):1–35, 2024.
- [21] M. Templ and J. Gonzalez-Rodriguez. Advancing forensic research: An examination of compositional data analysis with an application on petrol fraud detection. *Science & Justice*, 64(1):9–18, 2024.
- [22] Barbara Templ, Matthias Templ, Anouk N' Guyen van Chinh, and Urs Brändle. Leveraging sankey diagrams for enhanced curriculum planning in higher education. *The Curriculum Journal*, online first, 2024.

- [23] Johannes Gussenbauer, Matthias Templ, Siro Fritzmann, and Alexander Kowarik. Simulation of calibrated complex synthetic population data with xgboost. *Algorithms*, 17(6), 2024.
- [24] Caterina Gozzi, Matthias Templ, and Antonella Buccianti. Robust coda balances and the role of the variance in complex riverine geochemical systems. *Journal of Geochemical Exploration*, 259:107438, 2024.
- [25] Jürgen Degenfellner and Matthias Templ. Modeling bee hive dynamics: Assessing colony health using hive weight and environmental parameters. *Computers and Electronics in Agriculture*, 218:108742, 2024.
- [26] M. Templ. Enhancing precision in large-scale data analysis: An innovative robust imputation algorithm for managing outliers and missing values. *Mathematics*, 11(12), 2023.
- [27] Peter Filzmoser and Matthias Templ. Prof. Rudolf Dutter (1946-2023): Ein Nachruf. *Austrian Journal of Statistics*, 52(3):143–144, 2023.
- [28] Matthias Templ and Murat Sariyar. A systematic overview on methods to protect sensitive data provided for various analyses. *International Journal of Information Security*, August 2022.
- [29] M. Templ, C. Kanjala, and I. Siems. Privacy of study participants in open-access health and demographic surveillance system data: Requirements analysis for data anonymization. *JMIR Public Health Surveill*, 8(9):e34472, Sep 2022.
- [30] M. Templ, C. Gozzi, and A. Buccianti. A new version of the Langelier-Ludwig square diagram under a compositional perspective. *Journal of Geochemical Exploration*, 242:107084, 2022.
- [31] Matthias Templ and Barbara Templ. Statistical analysis of chemical element compositions in food science: Problems and possibilities. *Molecules*, 26(9):1–15, 2021.
- [32] Matthias Templ. Artificial neural networks to impute rounded zeros in compositional data. *Advances in Compositional Data Analysis*, Springer, 1, 2021.
- [33] Barbara Templ, Matthias Templ, Roberto Barbieri, Martin Meier, and V. Zufferey. Coincidence of temperature extremes and phenological events of grapevine varieties. *OENO One*, 55(1):367–383, 2021.
- [34] Sugnet Lubbe, Matthias Templ, and Peter Filzmoser. Comparison of zero replacement strategies for compositional data with large numbers of zeros. *Chemometrics and Intelligent Laboratory Systems*, pages 1–14, 2021.
- [35] K. G. van den Boogaart, P. Filzmoser, K. Hron, M. Templ, and R. Tolosana-Delgado. Classical and Robust Regression Analysis with Compositional Data. *Mathematical Geosciences*, October 2020.
- [36] Matthias Templ and Barbara Templ. Analysis of chemical compounds in beverages- guidance for establishing a compositional analysis. *Food Chemistry*, 325:1–7, 2020.
- [37] Matthias Templ and Christoph Heitz. Fleet management in free-floating bike sharing systems using predictive modelling and explorative tools. *Austrian Journal of Statistics*, 49(2):53–69, 2020.

- [38] Matthias Templ. Modeling and prediction of the impact factor of journals using open-access databases. *Austrian Journal of Statistics*, 49(5):35–58, 2020.
- [39] Stefan Sauermann, Chifundo Kanjala, and Matthias Templ. Preservation of individuals' privacy in shared covid-19 related data. *SSRN*, <http://dx.doi.org/10.2139/ssrn.3648430>:1–13, 2020.
- [40] Dedi Rosadi, Ezra Putranda Setiawan, Matthias Templ, and Peter Filzmoser. Robust covariance estimators for mean-variance portfolio optimization with transaction lots. *Operations Research Perspectives*, 7, 2020.
- [41] Matthias Templ, Johannes Gussenbauer, and Peter Filzmoser. Evaluation of robust outlier detection methods for zero-inflated complex data. *Journal of Applied Statistics*, 0(1):1–24, 2019.
- [42] Barbara Templ, Edina Mozes, Matthias Templ, Rita Foeldes, Adam Skirak, Andras Baldi, and Aniko Kovács-Hostyánszki. Habitat-dependency of transect walk and pan trap methods for bee sampling in farmlands. *Journal of Apicultural Science*, 63(1):93–115, 2019.
- [43] Bernhard Meindl and Matthias Templ. Feedback-based integration of the whole process of data anonymization in a graphical interface. *algorithms*, 12(191):1–20, 2019.
- [44] Ann-Kristin Kreuzmann, Soeren Pannier, Natalia Rojas, Timo Schmid, Matthias Templ, and Nikos Tzavidis. The R package emdi for the estimation and mapping of regional disaggregated indicators. *Journal of Statistical Software*, 91(7):1–33, 2019.
- [45] K Facevicova, Karel Hron, Valentin Todorov, and Matthias Templ. General approach to coordinate representation of compositional tables. *Scandinavian Journal of Statistics*, 45(4):879–899, 2018.
- [46] Matthias Templ, Bernhard Meindl, Alexander Kowarik, and Olivier Dupriez. Simulation of synthetic complex data: The R package simPop. *Journal of Statistical Software*, 79(10):1–38, 2017.
- [47] Matthias Templ, Karel Hron, and Peter Filzmoser. Exploratory tools for outlier detection in compositional data with structural zeros. *Journal of Applied Statistics*, 44(4):734–752, 2017.
- [48] Barbara Templ, Matthias Templ, Peter Filzmoser, Annamária Lehoczky, Bálint Czúcz, Eugenija Baksienė, Stefan Fleck, Hippla Gregow, Sabina Hodzic, Gunta Kalvane, Eero Kubin, Vello Palm, Visnja Romanovskaja, Visnja Vucetic, and Ana Zust. Phenological patterns of flowering across biogeographical regions of europe. *International Journal Of Biometeorology*, 61(7):1347–1358, 2017.
- [49] Jiajia Chen, Xiaoqin Zhang, Karel Hron, Matthias Templ, and Shengjia Li. Regression imputation with q-mode clustering for rounded zero replacement in high-dimensional compositional data. *Journal of Applied Statistics*, 45(11):2067–2080, 2017.
- [50] Matthias Templ, Karel Hron, Peter Filzmoser, and Alzbetka Gardlo. Imputation of rounded zeros for high-dimensional compositional data. *Chemometrics and Intelligent Laboratory Systems*, 155:183–190, 2016.
- [51] Barbara Templ, Stefan Fleck, and Matthias Templ. Change of plant phenophases explained by survival modelling. *International Journal Of Biometeorology*, 61(5):881–889, 2016.

- [52] Alexander Kowarik and Matthias Templ. Imputation with the R package VIM. *Journal of Statistical Software*, 74(7):1–16, 2016.
- [53] Johannes Huber, Bernhard Pötsch, Martina Gantschacher, and Matthias Templ. Routine treatment of cervical cytological cell changes. *Geburtshilfe Frauenheilkunde*, 76(10):1086–1091, 2016.
- [54] K Facevicova, Karel Hron, Valentin Todorov, and Matthias Templ. Compositional tables analysis in coordinates. *Scandinavian Journal of Statistics*, 43(4):962–977, 2016.
- [55] Gerald van den Boogaart, Raimon Tolosano-Delgado, and Matthias Templ. Regression with compositional response having unobserved components or below detection limit values. *Statistical Modelling*, 15(2):191–213, 2015.
- [56] Matthias Templ, Alexander Kowarik, and Bernhard Meindl. Statistical disclosure control for micro-data using the R package sdcMicro. *Journal of Statistical Software*, 67(4):1–36, 2015.
- [57] Matthias Templ. Quality indicators for statistical disclosure methods: A case study on the structure of earnings survey. *Journal of Official Statistics*, 31(4):737–761, 2015.
- [58] Josep Antoni Martin-Fernandez, Karel Hron, Matthias Templ, Peter Filzmoser, and Javier Palarea-Albaladejo. Bayesian-multiplicative treatment of count zeros in compositional data sets. *Statistical Modelling*, 15(2):134–158, 2015.
- [59] Alexander Kowarik, Bernhard Meindl, and Matthias Templ. sparktable: Generating graphical tables for websites and documents with r. *R Journal*, 7(1):24–37, 2015.
- [60] Peter Filzmoser, Karel Hron, Klara Hruzova, Alessandra Menafoglio, and Matthias Templ. Simplicial principal component analysis for density functions in Bayes spaces. *Computational Statistics and Data Analysis*, 94:330–350, 2015.
- [61] Juan Jose Egozcue, Vera Pawlowsky-Glahn, Matthias Templ, and Karel Hron. Independence in contingency tables using simplicial geometry. *Communications in Statistics - Theory and Methods*, 44(18):3978–3996, 2015.
- [62] Matthias Templ, Alexander Kowarik, and Bernhard Meindl. Development and current practice in using r at statistics austria. *Romanian Statistical Review*, 2:173–184, 2014.
- [63] Matthias Templ and Peter Filzmoser. Simulation and quality of a synthetic close-to-reality employer-employee population. *Journal of Applied Statistics*, 41(5):1053–1072, 2014.
- [64] Matthias Templ. Providing data with high utility and no disclosure risk for the public and researchers: An evaluation by advanced statistical disclosure risk methods. *Austrian Journal of Statistics*, 43(4):247–254, 2014.
- [65] Alexander Kowarik, Angelika Meraner, Matthias Templ, and Daniel Schopfhauser. Seasonal adjustment with the r-packages x12 and x12gui. *Journal of Statistical Software*, 62(2):1–21, 2014.
- [66] Karel Hron, Peter Filzmoser, Matthias Templ, et al. Robust regression with compositional response: Application to geosciences. In *Mathematics of Planet Earth, Lecture Notes in Earth System Sciences*, pages 87–90. Springer Verlag Berlin-Heidelberg, 2014.

- [67] K. Facevicova, Karel Hron, Valentin Todorov, D. Guo, and Matthias Templ. Logratio approach to statistical analysis of 2×2 compositional tables. *Journal of Applied Statistics*, 41:944–958, 2014.
- [68] Matthias Templ, Semagül Aklan, Peter Filzmoser, Matthias Preusser, and Johannes Hainfellner. Statistical indicators for the analysis of digitalized brain tumor images. *Austrian Journal of Statistics*, Volume 42(2):81–100, 2013.
- [69] Bernhard Meindl and Matthias Templ. Analysis of commercial and free and open source solvers for the cell suppression problem. *Transaction on Data Privacy*, Volume 6(2):147–159, 2013.
- [70] David Kepplinger, Matthias Templ, and S. Upadhyaya. Analysis of energy intensity in manufacturing industry using mixed-effects models. *Energy*, 59(15. September):754–763, 2013.
- [71] Karel Hron, Matthias Templ, and Peter Filzmoser. Estimation of a proportion in survey sampling using the logratio approach. *Metrika*, 76(6):799–818, 2013.
- [72] Andreas Alfons, Matthias Templ, and Peter Filzmoser. Robust estimation of economic indicators from survey samples based on pareto tail modelling. *Journal of the Royal Statistical Society: Series C (Applied Statistics)*, Vol. 62(Issue 2):271–286, 2013.
- [73] Andreas Alfons and Matthias Templ. Estimation of social exclusion indicators from complex surveys: The R package laeken. *Journal of Statistical Software*, Vol. 54(Issue 15, Sep 2013):1–25, 2013.
- [74] Matthias Templ and Valentin Todorov. Official statistics and survey methodology meets r: An overview of corresponding packages. *The Survey Statistician*, July(66):26–34, 2012.
- [75] Matthias Templ, Beat Hulliger, Alexander Kowarik, and Karin Fürst. Combining geographical information and traditional plots: the checkerplot. *International Journal of Geographical Information Science*, 1:3–16, 2012.
- [76] Matthias Templ, Andreas Alfons, and Peter Filzmoser. Exploring incomplete data using visualization techniques. *Advances in Data Analysis and Classification*, 6:29–47, 2012.
- [77] Matthias Templ. Correlation between indicators over time in thematic maps. *Austrian Journal of Statistics*, 41(1):67–79, 2012.
- [78] Josep Antoni Martin-Fernandez, Karel Hron, Matthias Templ, Peter Filzmoser, and Javier Palarea-Albaladejo. Model-based replacement of rounded zeros in compositional data: Classical and robust approaches. *Computational Statistics and Data Analysis*, 56:2688–2704, 2012.
- [79] Alexander Kowarik, Matthias Templ, Bernhard Meindl, Francois Fonteneau, and Bernd Prantner. Testing of ihsn c++ code and inclusion of new methods into sdcmicro. In Josep Domingo-Ferrer and Ilenia Tinnirello, editors, *Lecture Notes in Computer Science*, pages 63–77. Springer, Berlin, 2012.
- [80] Peter Filzmoser, Karel Hron, and Matthias Templ. Discriminant analysis for compositional data and robust parameter estimation. *Computational Statistics*, 27(4):585–604, 2012.

- [81] Valentin Todorov, Matthias Templ, and Peter Filzmoser. Detection of multivariate outliers in business survey data with incomplete information. *Advances in Data Analysis and Classification*, 5(1):37–56, 2011.
- [82] Matthias Templ, Alexander Kowarik, and Peter Filzmoser. Iterative stepwise regression imputation using standard and robust methods. *Computational Statistics and Data Analysis*, 55:2793–2806, 2011.
- [83] Matthias Templ, Karel Hron, and Peter Filzmoser. robCcompositions: an r-package for robust statistical analysis of compositional data. In *Compositional Data Analysis: Theory and Applications*, pages 341–355. John Wiley & Sons, 2011.
- [84] Gerlinde Dinges, Alexander Kowarik, Bernhard Meindl, and Matthias Templ. An open source approach for modern teaching methods: The interactive tgui system. *Journal of Statistical Software*, 39\7, 2011.
- [85] Gerlinde Dinges, Alexander Kowarik, Bernhard Meindl, and Matthias Templ. Moderne wege der wissensvermittlung: Statistikunterricht interaktiv. *Statistische Nachrichten*, 9:919–922, 2011.
- [86] Andreas Alfons, Matthias Templ, and Peter Filzmoser. Simulation of close-to-reality population data for household surveys with application to eu-silc. *Statistical Methods & Applications*, 20(3):383–407, 2011.
- [87] Matthias Templ and Bernhard Meindl. Practical applications in statistical disclosure control using r. In Jordi Nin and Javier Herranz, editors, *Privacy and Anonymity in Information Management Systems New Techniques for New Practical Problems*, pages 31–62. Springer Verlag, 2010.
- [88] Matthias Templ and Andreas Alfons. Disclosure risk of synthetic population data with application in the case of eu-silc. In Josep Domingo-Ferrer and Emmanouil Magkos, editors, *Privacy in Statistical Databases, Lecture Notes in Computer Science*, pages 174–186. Springer Verlag, Berlin, Heidelberg, 2010.
- [89] Karel Hron, Matthias Templ, and Peter Filzmoser. Imputation of missing values for compositional data using classical and robust methods. *Computational Statistics and Data Analysis*, 54:3095–3107, 2010.
- [90] Andreas Alfons, Matthias Templ, and Peter Filzmoser. An object-oriented framework for statistical simulation: The R package simFrame. *Journal of Statistical Software*, 37(3):1–36, 2010.
- [91] Matthias Templ and Thomas Petelin. A graphical user interface for microdata protection which provides reproducibility and interactions: the sdcmicro gui. *Transactions on Data Privacy*, 2:207–223, 2009.
- [92] Matthias Templ. Forschungsprojekt AMELI; regionale Schätzung von Armut und sozialem Zusammenhalt in Europa. *Statistische Nachrichten*, 2:173–177, 2009.
- [93] Matthias Templ. Clusteranalyse mit SPSS. mit faktorenanalyse. (book reviews). *Austrian Journal of Statistics*, 38:287–291, 2009.

- [94] Gerlinde Dinges and Matthias Templ. Motivation zur statistik - computergestützt lernen in der statistik austria. *Austrian Journal of Statistics*, 38(1):3–16, 2009.
- [95] Matthias Templ and Bernhard Meindl. Robustification of microdata masking methods and the comparison with existing methods. In Josep Domingo-Ferrer and Yücel Saygin, editors, *Lecture Notes on Computer Science*, page 14. Springer, Berlin/Heidelberg, 2008.
- [96] Matthias Templ and Bernhard Meindl. Robust statistics meets sdc: New disclosure risk measures for continuous microdata masking. In Josep Domingo-Ferrer and Yücel Saygin, editors, *Lecture Notes on Computer Science*, page 13. Springer, Berlin/Heidelberg, 2008.
- [97] Matthias Templ, Peter Filzmoser, and Clemens Reimann. Cluster analysis applied to regional geochemical data: Problems and possibilities. *Applied Geochemistry*, 23(8):2198–2213, 2008.
- [98] Matthias Templ. Statistical disclosure control for microdata using the r-package sdcmicro. *Transactions on Data Privacy*, 1:67–85, 2008.
- [99] Matthias Templ. Forschungsaktivitäten im Bereich statistischer Geheimhaltung von Mikrodaten. *Statistische Nachrichten*, 1:94–98, 2008.
- [100] Matthias Templ. Software development for SDC in R. In Josep Domingo-Ferrer and Luisa Franconi, editors, *Privacy in Statistical Databases, Lecture Notes in Computer Science, Vol. 4302*, pages 347–359. Springer, Berlin Heidelberg, 2006.

Published Interviews and Podcasts

- [101] Jeannie Wurz. Sharing data and protecting privacy in low- and middle-income countries, 2021. Interview with Matthias Templ and Chifundo Kanjala.
- [102] Jaennine Wuerz, Matthias Templ, and Chifundo Kanjala. Sharing data and protecting privacy in low- and middle-income countries. Art der nichttextlichen wissenschaftlichen Veröffentlichung: Interview, Projekt: ESTHER Switzerland, 2021.
- [103] Ernst Stadlober, Herwig Friedl, and Matthias Templ. Die theorie lebt in der praxis. ein interview mit ernst stadlober. *Austrian Journal of Statistics*, 45(2):69–80, 2016.
- [104] Ewald Kutzenberger, Werner Holzer, and Matthias Templ. Ein pakt mit den bürgern. ein interview mit ewald kutzenberger. *Austrian Journal of Statistics*, 44(1):47–57, 2015.
- [105] Peter Hackl, Werner Mueller, and Matthias Templ. Akademische und offizielle statistik vereint. ein interview mit peter hackl. *Austrian Journal of Statistics*, 44(3):85–95, 2015.
- [106] Wilfried Grossmann, Werner Mueller, and Matthias Templ. Ein interview mit wilfried grossmann. *Austrian Journal of Statistics*, 43(2):151–162, 2014.

Habilitation, PhD and master thesis, Supervisor, and Reviewer

- [107] C. Sluka. Anonymisation of clinical data for the swiss stroke registry. Master's thesis, ZHAW, Winterthur, Schweiz, 2023. supervisor: M. Templ.
- [108] Simon Wuersten. Implementation von Anonymisierungsmethoden für soziodemografische Mikrodaten. Master's thesis, Institut für Datenanalyse und Prozessdesign (ZHAW), 2021.
- [109] M. Hilti. Visualisierung des baumkatasters der stadt zürich. Master's thesis, ZHAW, Winterthur, Schweiz, 2021. supervisor: M. Templ.
- [110] Rainer Dangl. *Benchmarking in Cluster Analysis – Insights into Theory and Application written*. PhD thesis, Vienna University of Technology, 2021. Reviewer and examiner.
- [111] Andreas Blatt. Sampling dns traffic: A day in the life of the .at-zone. Master's thesis, Institute of Statistics and Mathematical Methods in Economics, 2020.
- [112] Juergen Degenfellner. Analysis of bee hive data using robust methods. Master's thesis, Institute of Statistics and Mathematical Methods in Economics, 2019.
- [113] Roberto Barbieri. Imputation of missing values: Deep learning vs. traditional methods. Master's thesis, Institute of Data Analysis and Process Design, ZHAW, 2019.
- [114] Mina Sohrabi. Statistical disclosure control: protection of confidential longitudinal data. Master's thesis, Institut für Stochastik und Wirtschaftsmathematik, 2018.
- [115] Gregor De Cilia. New computer-oriented teaching concepts in statistics. Master's thesis, Institut für Stochastik und Wirtschaftsmathematik, 2018.
- [116] Marlene Weinauer. Analysis and forecasting of austrian mortality data with methods of compositional data analysis. Master's thesis, Instituts für Stochastik und Wirtschaftsmathematik, 2017.
- [117] Thomas Ortner. *Local projections for high-dimensional data analysis*. PhD thesis, Vienna University of Technology, 2017. Reviewer.
- [118] Petra Klynlova. *Methodological contributions to compositional data analysis*. PhD thesis, Vienna University of Technology, 2016. Reviewer and examiner.
- [119] Klara Hruzova. *Economic applications of statistical analysis of compositional data*. PhD thesis, Department of Mathematical Analysis and Applications of Mathematics, Palacký university Olomouc, 2016.
- [120] Marius Totter. Disclosure risk estimation for survey data. Master's thesis, Institute of Statistics and Mathematical Methods in Economics, 2015.
- [121] Alexander Kowarik. *New Computational Tools and Methods for Official Statistics*. PhD thesis, Institute of Statistics and Mathematical Methods in Economics, 2015.

- [122] Johannes Gussenbauer. Robust statistical methods for outlier detection with application to household expenditure data. Master's thesis, Institute of Statistics & Mathematical Methods in Economics, 2015.
- [123] Verena Puchner. Evaluierung von methoden des statistischen matchings und ausgewählte sae methoden mit anwendung auf mikrozensus und eu-silc. Master's thesis, E107 Institut für Statistik und Wahrscheinlichkeitstheorie, 2013.
- [124] Matthias Templ. Methodological and computational developments for pre-processing and dissemination of survey microdata. Habilitationsschrift, 2012.
- [125] Karoline Geißler. The synthetic close-to-reality employer-employee population: Simulation and benchmarking statistics. Master's thesis, E107 Institut fuer Statistik und Wahrscheinlichkeitstheorie, 2012.
- [126] Semagül Aklan. Development and implementation of statistical indicators for the assessment of brain tumors. Master's thesis, E107, 2012.
- [127] Stefan Zechner. Visualization of indicators in R with application to EU-SILC. Master's thesis, Institute of Statistics and Mathematical Methods in Economic, Vienna University of Technology, 2010.
- [128] Angelika Meraner. Outlier detection for semi-continuous variables. Master's thesis, E107, 2010.
- [129] Matthias Templ. *New Developments in Statistical Disclosure Control and Imputation in Official Statistics*. PhD thesis, E107, 2009.
- [130] Matthias Templ. Cluster analysis with application to data from geochemistry. Master's thesis, E107, 2003.

Proceedings

- [131] Oscar Thees, Jiří Novák, and Matthias Templ. Evaluation of synthetic data generators on complex tabular data. In Josep Domingo-Ferrer and Melek Önen, editors, *Privacy in Statistical Databases*, pages 194–209, Cham, 2024. Springer Nature Switzerland.
- [132] Matthias Templ. Can the compositional nature of compositional data be ignored by using deep learning approaches? In Nicola Salvati, Cira Perna, Stefano Marchetti, and Raymond Chambers, editors, *Studies in Theoretical and Applied Statistics*, pages 151–166, Cham, 2022. Springer International Publishing.
- [133] Matthias Templ. Can we ignore the compositional nature of compositional data by using deep learning approaches? In Cira Perna, Nicola Salvati, and Francesco Schirripa Spagnolo, editors, *Book of Short Papers. SIS 2021*, volume 1, pages 243–248, London, 2021. SIS, Pearson.
- [134] M. Templ. Can we ignore the compositional nature of compositional data by using deep learning approaches? : Possiamo ignorare la natura compositazionale dei dati composizionali usando gli approcci di deep learning? In Cira Perna, Nicola Salvati, and Francesco Schirripa Spagnolo,

editors, *Book of Short Papers SIS 2021*, pages 243–248, London, 2021. Pearson. Surface Inspection Summit Europe, Aachen, Germany, 9-10 November 2021.

- [135] Christoph Heitz, Roman Etschmann, Raul Stöckle, Thomas Bachmann, and Matthias Templ. User-based redistribution in free-floating bike sharing systems. In *Operations Research Proceedings 2018*, Heidelberg, Germany, 2019. Annual International Conference of the German Operations Research Society (GOR), Springer Nature Switzerland AG.
- [136] Ann-Kristin Kreuzmann, Soeren Pannier, Natalia Rojas, Timo Schmid, Matthias Templ, and Nikos Tzavidis. Small area estimation in r with application to mexican income data. In *NTTS 2017 Proceedings*, pages 1–5, 2017.
- [137] Matthias Templ. Synthetic data: Modelling and generating complex close-to-reality data for public use. In Eva Fiserová, editor, *Olomoucian Days of Applied Mathematics*, page 81, Olomouc, 2015. Univerzita Palackeho Olomouc, Univerzita Palackeho Olomouc.
- [138] Matthias Templ and Bernhard Meindl. Methods and tools for the simulation of synthetic populations (in session: Statistical modelling with r). In *CFE-ERCIM 2014 - Book of Abstracts*, page ##, 2014.
- [139] Matthias Templ and Bernhard Meindl. Methods and tools for the generation of synthetic populations. a brief review. In Josep Domingo-Ferrer, editor, *Proceedings of the Privacy in Statistical Databases 2014, UNESCO Chair in Data Privacy, International Conference, PSD 2014, Ibiza, Spain, September 17-19, 2014*. Springer Verlag, 2014.
- [140] Matthias Templ. A methodological and computational framework for statistical disclosure control. In *Conference of European Statistics Stakeholders. Methodologists, Producers and Users of European Statistics*, page 36. Sapienza University of Rome, 2014.
- [141] Matthias Templ. Contributions to robustness and outlier identification (chair - session es140). In *CFE-ERCIM 2014 - Book of Abstracts*, page ##, 2014.
- [142] Alexander Kowarik, Matthias Templ, and Daniel Schopfhauser. New features of vim - visualisation and imputation of missing values. In *UNECE Worksession on Statistical Data Editing*, Paris, 2014. UNECE, UNECE.
- [143] Alexander Kowarik, Bernhard Meindl, and Matthias Templ. sparktable: Generating graphical tables for websites and documents with r. In *Proceedings of the European Conference on Quality in Official Statistics 2014*, Wien, 2014. Statistik Austria.
- [144] Karel Hron, Matthias Templ, and Valentin Todorov. Robust exploratory data analysis of compositional tables (in session: Robust and nonparametric methods). In *CFE-ERCIM 2014 - Book of Abstracts*, page ##, 2014.
- [145] Karel Hron, Alessandra Menafoglio, Matthias Templ, Klara Hruzova, and Peter Filzmoser. Simplicial principal component analysis for density functions in Bayes spaces. In *CFE-ERCIM 2014 - Book of Abstracts*, 2014.
- [146] Daniel Elias, Birgit Nadler, Friedrich Nadler, Alexander Dinkel, Stefan Gürtler, Alexander Kowarik, Bernhard Meindl, and Matthias Templ. Quatra (software and services for the quality management of traffic data). In *Transport Research Arena (TRA) 2014 Proceedings /*

5th Conference: Transport Solutions from Research to Deployment, pages 1–10. European Commission, Transport Research Arena (TRA), 2014.

- [147] Judith Zehetgruber, Peter Filzmoser, Karel Hron, Matthias Templ, et al. Robust multivariate regression with compositional data. In Karel Hron, Peter Filzmoser, and Matthias Templ, editors, *Proceedings of the 5th International Workshop on Compositional Data Analysis CoDaWork 2013 June 3-7, 2013, Vorau, Austria*, page 65, 2013.
- [148] G van den Boogaart, Raimon Tolosana-Delgado, Karel Hron, Matthias Templ, and Peter Filzmoser. Compositional regression with unobserved components or below detection limit values. In Karel Hron, Peter Filzmoser, and Matthias Templ, editors, *Proceedings of the 5th International Workshop on Compositional Data Analysis CoDaWork 2013 June 3-7, 2013, Vorau, Austria*, page 59, 2013.
- [149] Valentin Todorov, K Facevicova, Karel Hron, D. Guo, and Matthias Templ. Statistical analysis of compositional 2x2 tables with an economic application. In Karel Hron, Peter Filzmoser, and Matthias Templ, editors, *Proceedings of the 5th International Workshop on Compositional Data Analysis CoDaWork 2013 June 3-7, 2013, Vorau, Austria*, page 58, 2013.
- [150] Matthias Templ, G van den Boogaart, Jiri Eichler, Peter Filzmoser, Karel Hron, and Raimon Tolosana-Delgado. Software compositionsgui. In Karel Hron, Peter Filzmoser, and Matthias Templ, editors, *Proceedings of the 5th International Workshop on Compositional Data Analysis CoDaWork 2013 June 3-7, 2013, Vorau, Austria*, page ##, 2013.
- [151] Matthias Templ, Karel Hron, Peter Filzmoser, and Gianna Monti. Methods to detect outliers in compositional data with structural zeros. In Karel Hron, Peter Filzmoser, and Matthias Templ, editors, *Proceedings of the 5th International Workshop on Compositional Data Analysis CoDaWork 2013 June 3-7, 2013, Vorau, Austria*, page 56, 2013.
- [152] Matthias Templ, Karel Hron, Peter Filzmoser, et al. The R packages robCompositions and compositionsGUI. In Karel Hron, Peter Filzmoser, and Matthias Templ, editors, *Proceedings of the 5th International Workshop on Compositional Data Analysis CoDaWork 2013 June 3-7, 2013, Vorau, Austria*, page ##, 2013.
- [153] Matthias Templ. Supporting tools. In Karel Hron, Peter Filzmoser, and Matthias Templ, editors, *Proceedings of the 5th International Workshop on Compositional Data Analysis CoDaWork 2013 June 3-7, 2013, Vorau, Austria*, page ##, 2013.
- [154] Matthias Templ. Providing data with high utility and no disclosure risk for the public and researchers - an evaluation by advanced statistical disclosure risk methods. In *Proceedings of the 10th International Conference Computer Data Analysis & Modelling 2013*, pages 57–66. Belarusian State University, 2013.
- [155] Gianna Monti, Karel Hron, Peter Filzmoser, and Matthias Templ. Covariance-based outlier detection for compositional data with structural zeros: Application to italian survey of household income and wealth data. In Di Vita e Pensiero, editor, *Dipartimento di Economia, Metodi Quantitativi e Strategie di Impresa*. Vita e Pensiero, Vita e Pensiero, 2013.
- [156] Alexander Kowarik and Matthias Templ. Make your data confidential with the sdcmicro and sdcmicrogui packages. In *The R User Conference, useR! 2013, Book of Contributed Abstracts*, page 64, Albacete, Spain, 2013. University of Castilla-La Mancha.

- [157] Matthias Templ and Valentin Todorov. Screening methods and tools for the unido industrial statistics (indstat). In UNECE, editor, *Conference of European Statisticians, Norway/Oslo, September 2012*, pages 1–19, 2012.
- [158] Matthias Templ and Valentin Todorov. Official statistics and survey methodology meets r: An overview of corresponding packages. In *Proceedings of the International Conference on Establishment Surveys*, 2012.
- [159] Matthias Templ, Alexander Kowarik, and Beat Hulliger. Visualization of complex statistical information in maps. In *2012 JSM Proceedings*. American Statistical Association, 2012.
- [160] Matthias Templ, Beat Hulliger, and Alexander Kowarik. Visualization of regional indicators with the checkerplot. In Peter Filzmoser, editor, *Book of Abstracts*, page 141, 2012.
- [161] Josep Antoni Martin-Fernandez, Karel Hron, Matthias Templ, Peter Filzmoser, and Javier Palarea-Albaladejo. Evaluating the performance of a Bayesian-multiplicative treatment of zeros in compositional data sets. In *Proceedings of COMPSTAT2012*, page ##, 2012.
- [162] Alexander Kowarik, Angelika Meraner, Daniel Schopfhauser, and Matthias Templ. Interactive adjustment and outlier detection of time dependent data in r. In UNECE, editor, *Conference of European Statisticians, Norway/Oslo, September 2012*, pages 1–11, 2012.
- [163] Valentin Todorov, Matthias Templ, and Peter Filzmoser. Software for multivariate outlier detection in survey data. In *Proceedings of the Conference of European Statisticians, Work Session on Statistical Data Editing*, 2011.
- [164] Matthias Templ, Alexander Kowarik, and Peter Filzmoser. Imputation of complex data with r-package vim: traditional and new methods based on robust estimation. In *Proceedings of the Conference of European Statisticians, Work Session on Statistical Data Editing*, 2011.
- [165] Matthias Templ, Peter Filzmoser, and Karel Hron. Analysis of compositional data using robust methods. the r-package robCcompositions. In *Proceedings of the Conference of European Statisticians, Work Session on Statistical Data Editing*, 2011.
- [166] Matthias Templ, Andreas Alfons, Alexander Kowarik, Bernhard Meindl, Peter Filzmoser, Beat Hulliger, and Daniela Lussmann. Visualisation of indicators within the ameli project. In *Proceedings of the NTTS 2011 conference*, 2011.
- [167] Matthias Templ, Andreas Alfons, Peter Filzmoser, and Josef Holzer. Estimation of social inclusion indicators with bounded influence of incomes by tail modeling. In *Book of Abstracts of the Olomoucian Days of Applied Mathematics*, page 61, 2011.
- [168] Matthias Templ, Andreas Alfons, and Peter Filzmoser. Robust semi-parametric estimation of social cohesion and poverty indicators. In *Proceedings of the 58th ISI Conference in Dublin*. International Statistical Institute, 2011.
- [169] Matthias Templ. Comparison of perturbation methods based on pre-defined quality indicators. In *Online Proceedings of the Conference of European Statisticians*, 2011.
- [170] Karel Hron, Matthias Templ, and Peter Filzmoser. Classical and robust imputation of missing values for compositional data using balances. In *Proceedings of the 4th International Workshop on Compositional Data Analysis*, 2011.

- [171] Karel Hron, Eva Fiserová, Peter Filzmoser, and Matthias Templ. On the interpretation of orthonormal coordinates for compositional data with applications. In *Book of Abstracts of the Olomoucian Days of Applied Mathematics*, page 27, 2011.
- [172] Peter Filzmoser, Karel Hron, and Matthias Templ. Robust compositional data analysis. In *Proceedings of the 4th International Workshop on Compositional Data Analysis*, page 4, 2011.
- [173] Peter Filzmoser, Karel Hron, and Matthias Templ. Discriminant analysis for compositional data and robust parameter estimation. In *Book of Abstracts of the Olomoucian Days of Applied Mathematics*, page 17, 2011.
- [174] Andreas Alfons, Matthias Templ, and Peter Filzmoser. Robust estimation of social inclusion indicators based on pareto tail modeling. In *Proceedings of the NTTS 2011 conference*, 2011.
- [175] Matthias Templ, A. Kowarik, Peter Filzmoser, and Andreas Alfons. Iterative imputation of complex survey data using robust methods. In *Abstracts of the International Conference on Indicators and Survey Methodology*, page 51, 2010.
- [176] Matthias Templ, Karel Hron, and Peter Filzmoser. robCcompositions: An R-package for robust statistical analysis of compositional data. In *Abstracts of Contributed Presentations*, page 161, 2010.
- [177] Beat Hulliger, Daniela Lussmann, and Matthias Templ. Visualisation in the ameli project. In *Abstracts of the Workshop on Exploratory Data Analysis and Visualization*, page 20, 2010.
- [178] Karel Hron, Matthias Templ, and Peter Filzmoser. Exploratory compositional data analysis using the r-package robCcompositions. In S. Aivazian, Peter Filzmoser, and Yuriy Kharin, editors, *Proceedings of the Ninth International Conference Data Analysis and Modeling*, pages 179–186, 1, 2010.
- [179] Karel Hron, Peter Filzmoser, and Matthias Templ. Classical and robust simple random sampling for compositional data. In *Abstracts of the International Conference on Indicators and Survey Methodology*, page 33, 2010.
- [180] Andreas Alfons, Matthias Templ, Peter Filzmoser, Josef Holzer, and Gil Gonzalez-Rodriguez. A comparison of robust methods for pareto tail modeling in the case of laeken indicators. In Christian Borgelt, Gil Gonzalez-Rodriguez, W. Trutschnig, M.A. Lubiano, M.A. Gil, P. Grzegorzewski, and Olgierd Hryniewicz, editors, *Combining Soft Computing and Statistical Methods in Data Analysis*, pages 17–24, 2010.
- [181] Andreas Alfons, Matthias Templ, and Peter Filzmoser. The R package: simFrame: An object-oriented approach towards simulation studies in statistics. In *Abstracts of Contributed Presentations*, page 13, 2010.
- [182] Andreas Alfons, Matthias Templ, and Peter Filzmoser. Exploring microdata with missing information. In *Abstracts of the Workshop on Exploratory Data Analysis and Visualization*, page 11, 2010.
- [183] Andreas Alfons, Matthias Templ, and Peter Filzmoser. Contamination models in the R package simFrame for statistical simulation. In S. Aivazian, Peter Filzmoser, and Yuriy Kharin,

editors, *Proceedings of the Ninth International Conference on Computer Data Analysis and Modeling*, pages 178–181, 2, 2010.

- [184] Andreas Alfons, Stefan Kraft, Matthias Templ, and Peter Filzmoser. Simulation of population data for complex household surveys. In *Abstracts of the International Conference on Indicators and Survey Methodology*, page 13, 2010.
- [185] Valentin Todorov, Matthias Templ, and Peter Filzmoser. Robust outlier detection in survey data. In *Abstracts of the International Conference in Robust Statistics*, pages 148–149, 2009.
- [186] Valentin Todorov, Matthias Templ, and Peter Filzmoser. Outlier detection in survey data using robust methods. In *Proceedings of the Conference of European Statisticians*, 2009.
- [187] Matthias Templ, Alexander Kowarik, and Peter Filzmoser. Iterative robust model-based imputation of complex data. In *Abstracts of the Third International Conference on Computational and Financial Econometrics (CFE 09) and Second Workshop of the ERCIM Working Group on Computing & Statistics (ERCIM) 09*, page 29, 2009.
- [188] Matthias Templ, Peter Filzmoser, and Karel Hron. Robustness for compositional data using the R package: robCcompositions. In *Abstracts of the International Conference in Robust Statistics*, pages 146–147, 2009.
- [189] Matthias Templ, Peter Filzmoser, and Karel Hron. Robust imputation of missing values in compositional data using the r-package robCcompositions. In *Proceedings of the NTTS Conference*, 2009.
- [190] Matthias Templ, Peter Filzmoser, and Karel Hron. Missing values in compositional data: Imputation and diagnostics. In *Abstracts of the 6th International Conference on Computational Management Science*, page 16, 2009.
- [191] Matthias Templ, Peter Filzmoser, and Karel Hron. Imputation of item non-responses in compositional data using robust methods. In *Proceedings of the Conference of European Statisticians*, Neuchatel, 2009.
- [192] Matthias Templ, Andreas Alfons, and Peter Filzmoser. Exploring the multivariate structure of missing values using the R package VIM. In *Abstracts of the 5th R useR Conference*, page 194, Rennes, 2009.
- [193] Matthias Templ and Andreas Alfons. Visualisierung von fehlenden werten. In *Tagungsband zum 18. Wissenschaftlichen Kolloquium "Informationsvisualisierung - Grafische Aufbereitung und Analyse von statistischen Daten"*. Statistisches Bundesamt Deutschland, 2009.
- [194] Matthias Templ. An interactive graphical user interface for microdata protection which allows reproducibility. In *Abstracts of the conference of european statisticians*, 2009.
- [195] Peter Filzmoser, Heinrich Fritz, Karel Horn, and Matthias Templ. The estimation of missing data in presence of outliers: computational aspects. In *Abstracts of the Third International Conference on Computational and Financial Econometrics (CFE 09) and Second Workshop of the ERCIM Working Group on Computing & Statistics (ERCIM) 09*, page 29, 2009.

- [196] Andreas Alfons, Matthias Templ, and Peter Filzmoser. On the influence of imputation methods on laeken indicators: simulations and recommendations. In *Proceedings of the Conference of European Statisticians*, Neuchatel, 2009.
- [197] Andreas Alfons and Matthias Templ. Simulation in robust statistics using the R package simFrame. In *Abstracts of the Third International Conference on Computational and Financial Econometrics (CFE 09) and Second Workshop of the ERCIM Working Group on Computing & Statistics (ERCIM 09)*, page 89, 2009.
- [198] Valentin Todorov, Matthias Templ, and Peter Filzmoser. Outlier detection in survey data. In *Proceedings in Computational Statistics*. Physica-Verlag, 2008.
- [199] Matthias Templ. Visualisation of missing values before imputation. In *Abstracts of the European Conference on Quality in Official Statistics*, pages 49–50, 2008.
- [200] Matthias Templ. Distance-based disclosure risk for record linkage in official statistics. In *Abstracts of the International Conference On Robust Statistics 2008*, page 97, 2008.
- [201] Karel Hron, Matthias Templ, and Peter Filzmoser. A robust approach for dealing with missing values in compositional data. In *Abstracts of the International Conference on Robust Statistics 2008*, pages 48–49, 2008.
- [202] Andreas Alfons, Stefan Kraft, Matthias Templ, and Peter Filzmoser. Analyse und Visualisierung von EU-SILC Daten in R. In *Abstracts of the Workshop TU Wien/TU Dresden*, page 6, 2008.
- [203] Matthias Templ. sdcmicro: A package for statistical disclosure control in r. In *Bulletin of the International Statistical Institute, 56th Session*, Lissabon, 2007. ISI 2007.
- [204] Matthias Templ. sdcmicro: A new flexible r-package for the generation of anonymised microdata - design issues and new methods. In *Proceedings of the Joint UNECE/Eurostat work session on statistical data confidentiality*, 2007.
- [205] Bernhard Meindl and Matthias Templ. The anonymisation of the cvts2 and income tax dataset. an approach using r-package "sdc micro". In *Proceedings of the Joint UNECE/Eurostat work session on statistical data confidentiality*, 2007.
- [206] Matthias Templ. Experimental r-packages for data disclosure. In *Proceedings of the European Conference on Quality in Survey Statistics*, Cardiff, UK, 2006. European Conference on Quality in Survey Statistics.

Research Reports

- [207] Martina Gantschacher, Doris Ossberger, Matthias Templ, Christa Wirthumer-Hoche, and Markus Wolf. *Forschungsbericht: Studienbericht: Sprechende Gebrauchs- und Produktinformation*. Schriftenreihe Gesundheitsreihe und -wissenschaft. Sigmund Freud PrivatUniversitäts Verlag, Wien, 2013.
- [208] Matthias Templ. Artificial neural networks to impute rounded zeros in compositional data. Technical Report arXiv:2012.10300, E105 - Institut für Stochastik und Wirtschaftsmathematik; Technische Universität Wien, 2020.

- [209] Alexander Kowarik, Bernhard Meindl, and Matthias Templ. Generating a synthetic population data set for mauritania. Technical Report 1, E105 - Institut für Stochastik und Wirtschaftsmathematik; Technische Universität Wien, 2016.
- [210] Peter Filzmoser, Matthias Templ, and Johannes Gussenbauer. Short overview on outlier detection methods. Technical Report Deliverable 1 Contract with world bank (1157976), E105 - Institut für Stochastik und Wirtschaftsmathematik; Technische Universität Wien, 2015.
- [211] Peter Filzmoser, Johannes Gussenbauer, and Matthias Templ. Detecting outliers in household consumption survey data. Technical Report Deliverable 4. Contract with world bank (1157976), E105 - Institut für Stochastik und Wirtschaftsmathematik; Technische Universität Wien, 2015.
- [212] Matthias Templ, Bernhard Meindl, Alexander Kowarik, and Shuang Chen. Introduction to statistical disclosure control (sdc). Technical Report IHSN Working Paper No 007, E105 - Institut für Stochastik und Wirtschaftsmathematik; Technische Universität Wien, 2014.
- [213] Matthias Templ, Bernhard Meindl, and Alexander Kowarik. Tutorial for sdcmicrogui. Technical report, E105 - Institut für Stochastik und Wirtschaftsmathematik; Technische Universität Wien, 2014.
- [214] Matthias Templ, Alexander Kowarik, and Bernhard Meindl. Statistical disclosure control methods for anonymization of microdata and risk estimation. Technical report, E105 - Institut für Stochastik und Wirtschaftsmathematik; Technische Universität Wien, 2014.
- [215] Alexander Kowarik, Matthias Templ, Bernhard Meindl, and Francois Fontenau. Graphical user interface for package sdcMicro. Technical report, E105 - Institut für Stochastik und Wirtschaftsmathematik; Technische Universität Wien, 2014.
- [216] Karel Hron, Alessandra Menafoglio, Matthias Templ, Klara Hruzova, and Peter Filzmoser. Simplicial principal component analysis for density functions in Bayes spaces. Technical Report MOX-report 25/2014, E105 - Institut für Stochastik und Wirtschaftsmathematik; Technische Universität Wien, 2014.
- [217] Valentin Todorov and Matthias Templ. R in the statistical office: Part ii. Technical report, E105 - Institut für Stochastik und Wirtschaftsmathematik; Technische Universität Wien, 2012.
- [218] Bernhard Meindl and Matthias Templ. Analysis of commercial and free and open source solvers for linear optimization problems. Technical Report CS-2012-1, E105 - Institut für Stochastik und Wirtschaftsmathematik; Technische Universität Wien, 2012.
- [219] Andreas Alfons and Matthias Templ. Estimation of social exclusion indicators from complex surveys: The R package laeken. Technical report, E105 - Institut für Stochastik und Wirtschaftsmathematik; Technische Universität Wien, 2012.
- [220] Matthias Templ, Alexander Kowarik, and Peter Filzmoser. Em-based regression imputation using robust methods. Technical Report Europ. Commission, FP7-SSH-2007-21732, WP4-D4.2-8, E105 - Institut für Stochastik und Wirtschaftsmathematik; Technische Universität Wien, 2011.
- [221] Matthias Templ, Beat Hulliger, Daniela Lussmann, and Alexander Kowarik. The checkerplot. Technical Report Europ. Commission, FP7-SSH-2007-21732, WP8-D8.2-5, E105 - Institut für Stochastik und Wirtschaftsmathematik; Technische Universität Wien, 2011.

- [222] Matthias Templ, Beat Hulliger, Andreas Alfons, Daniela Lussmann, and Peter Filzmoser. Visualisation tools. Technical Report Europ. Commission, FP7-SSH-2007-21732, WP8-D8.2, E105 - Institut für Stochastik und Wirtschaftsmathematik; Technische Universität Wien, 2011.
- [223] Matthias Templ, Karel Hron, and Peter Filzmoser. Robust imputation for compositional data. Technical Report Europ. Commission, FP7-SSH-2007-21732, WP4-D4.2-11, E105 - Institut für Stochastik und Wirtschaftsmathematik; Technische Universität Wien, 2011.
- [224] Matthias Templ, Andreas Alfons, Monique Graf, Beat Hulliger, Risto Lehtonen, Tobias Schoch, Ari Veijanen, and Stefan Zins. References and description of R packages. Technical Report Europ. Commission, FP7-SSH-2007-21732, WP10-D10.3-1, E105 - Institut für Stochastik und Wirtschaftsmathematik; Technische Universität Wien, 2011.
- [225] Matthias Templ, Andreas Alfons, Peter Filzmoser, Monique Graf, Beat Hulliger, Jan-Philipp Kolb, Risto Lehtonen, Ralf Münnich, Desislava Nedyalkova, Tobias Schoch, Ari Veijanen, and Stefan Zins. R packages plus manual. Technical Report Europ. Commission, FP7-SSH-2007-21732, WP10-D10.3, E105 - Institut für Stochastik und Wirtschaftsmathematik; Technische Universität Wien, 2011.
- [226] Matthias Templ, Andreas Alfons, and Peter Filzmoser. Diagnostic tools for missing values. Technical Report Europ. Commission, FP7-SSH-2007-21732, WP8-D8.2-8, E105 - Institut für Stochastik und Wirtschaftsmathematik; Technische Universität Wien, 2011.
- [227] Matthias Templ and Andreas Alfons. Variance estimation of indicators using package laeken. Technical Report Europ. Commission, FP7-SSH-2007-21732, WP10-D10.3-7, E105 - Institut für Stochastik und Wirtschaftsmathematik; Technische Universität Wien, 2011.
- [228] Matthias Templ and Andreas Alfons. Variance estimation of indicators on social exclusion and poverty using the R package: laeken. Technical Report CS-2011-3, E105 - Institut für Stochastik und Wirtschaftsmathematik; Technische Universität Wien, 2011.
- [229] Matthias Templ and Andreas Alfons. Standard methods for social exclusion indicators in package. Technical Report Europ. Commission, FP7-SSH-2007-21732, WP10-D10.3-5, E105 - Institut für Stochastik und Wirtschaftsmathematik; Technische Universität Wien, 2011.
- [230] Matthias Templ and Andreas Alfons. Standard methods for point estimation of indicators on social exclusion and poverty using the R package: laeken. Technical Report CS-2011-1, E105 - Institut für Stochastik und Wirtschaftsmathematik; Technische Universität Wien, 2011.
- [231] Matthias Templ and Andreas Alfons. An application of vim to eu-silc data. Technical Report Europ. Commission, FP7-SSH-2007-21732, WP10-D10.3-4, E105 - Institut für Stochastik und Wirtschaftsmathematik; Technische Universität Wien, 2011.
- [232] Matthias Templ. Estimators and model predictions from the structural earnings survey for benchmarking statistical disclosure methods. Technical Report CS-2011-4, E105 - Institut für Stochastik und Wirtschaftsmathematik; Technische Universität Wien, 2011.
- [233] Ralf Munnich, Stefan Zins, Andreas Alfons, Christian Bruch, Peter Filzmoser, Monique Graf, Beat Hulliger, Jan-Philipp Kolb, Risto Lehtonen, Daniela Lussmann, Angelika Meraner, Mikko Myrskylä, Desislava Nedyalkova, Tobias Schoch, Matthias Templ, Maria Valaste,

- and Ari Veijanen. Policy recommendations and methodological report. Technical Report Europ. Commission,FP7-SSH-2007-217322,WP6-D10-1, E105 - Institut für Stochastik und Wirtschaftsmathematik; Technische Universität Wien, 2011.
- [234] Angelika Meraner, Peter Filzmoser, and Matthias Templ. Robust methods for semi-continuous data. Technical Report Europ. Commission,FP7-SSH-2007-21732,WP4-D4.2-12, E105 - Institut für Stochastik und Wirtschaftsmathematik; Technische Universität Wien, 2011.
- [235] Daniela Lussmann, Beat Hulliger, and Matthias Templ. State-of-the-art in visualisation of indicators. Technical Report Europ. Commission,FP7-SSH-2007-21732,WP8-D8.2-2, E105 - Institut für Stochastik und Wirtschaftsmathematik; Technische Universität Wien, 2011.
- [236] Alexander Kowarik, Bernhard Meindl, Matthias Templ, and Stefan Zechner. Applications of sparktables. Technical Report Europ. Commission,FP7-SSH-2007-21732,WP8-D8.2-6, E105 - Institut für Stochastik und Wirtschaftsmathematik; Technische Universität Wien, 2011.
- [237] Beat Hulliger, Stefan Zechner, Matthias Templ, and Peter Filzmoser. Evaluation plots: Revision and extensions. Technical Report Europ. Commission,FP7-SSH-2007-21732,WP8-D8.2-3, E105 - Institut für Stochastik und Wirtschaftsmathematik; Technische Universität Wien, 2011.
- [238] Beat Hulliger, Tobias Schoch, Andreas Alfons, Josef Holzer, Peter Filzmoser, Angelika Meraner, and Matthias Templ. Wp4: Robustness. Technical Report Europ. Commission,FP7-SSH-2007-21732,WP7-D7.1-10, E105 - Institut für Stochastik und Wirtschaftsmathematik; Technische Universität Wien, 2011.
- [239] Beat Hulliger, Andreas Alfons, Peter Filzmoser, Angelika Meraner, Tobias Schoch, and Matthias Templ. Robust methodology for laeken indicators. Technical Report Europ. Commission,FP7-SSH-2007-21732,WP4-D4.2, E105 - Institut für Stochastik und Wirtschaftsmathematik; Technische Universität Wien, 2011.
- [240] Beat Hulliger, Andreas Alfons, Peter Filzmoser, Angelika Meraner, Tobias Schoch, and Matthias Templ. R programmes for robust procedures including manual. Technical Report Europ. Commission,FP7-SSH-2007-21732,WP4-D4.1, E105 - Institut für Stochastik und Wirtschaftsmathematik; Technische Universität Wien, 2011.
- [241] Beat Hulliger, Andreas Alfons, Christian Bruch, Peter Filzmoser, Monique Graf, Jan-Philipp Kolb, Risto Lehtonen, Daniela Lussmann, Angelika Meraner, Ralf Münnich, Mikko Myrskylä, Desislava Nedyalkova, Tobias Schoch, Matthias Templ, Maria Valaste, Ari Veijanen, and Stefan Zins. Report on the simulation results: Appendix. Technical Report Europ. Commission,FP7-SSH-2007-21732,WP7-D7.1-App, E105 - Institut für Stochastik und Wirtschaftsmathematik; Technische Universität Wien, 2011.
- [242] Beat Hulliger, Andreas Alfons, Christian Bruch, Peter Filzmoser, Monique Graf, Jan-Philipp Kolb, Risto Lehtonen, Daniela Lussmann, Angelika Meraner, Ralf Münnich, Mikko Myrskylä, Desislava Nedyalkova, Tobias Schoch, Matthias Templ, Maria Valaste, Ari Veijanen, and Stefan Zins. Report on the simulation results. Technical Report Europ. Commission,FP7-SSH-2007-21732,WP7-D7.1, E105 - Institut für Stochastik und Wirtschaftsmathematik; Technische Universität Wien, 2011.

- [243] Monique Graf, Andreas Alfons, Christian Bruch, Peter Filzmoser, Beat Hulliger, Risto Lehtonen, Bernhard Meindl, Ralf Münnich, Tobias Schoch, Matthias Templ, Maria Valaste, Ariane Wenger, and Stefan Zins. State-of-the-art of indicators on poverty and social exclusion the laeken indicators. Technical Report Europ. Commission, FP7-SSH-2007-21732, WP1-D1.1, E105 - Institut für Stochastik und Wirtschaftsmathematik; Technische Universität Wien, 2011.
- [244] Andreas Alfons, Matthias Templ, Peter Filzmoser, and Josef Holzer. Semi-parametric robust estimation. Technical Report Europ. Commission, FP7-SSH-2007-21732, WP4-D4.2-3, E105 - Institut für Stochastik und Wirtschaftsmathematik; Technische Universität Wien, 2011.
- [245] Andreas Alfons, Matthias Templ, Peter Filzmoser, and Josef Holzer. Robust pareto tail modeling with package laeken. Technical Report Europ. Commission, FP7-SSH-2007-21732, WP10-D10.3-6, E105 - Institut für Stochastik und Wirtschaftsmathematik; Technische Universität Wien, 2011.
- [246] Andreas Alfons, Matthias Templ, Peter Filzmoser, and Josef Holzer. Robust pareto tail modeling for the estimation of indicators on social exclusion using the R package: laeken. Technical Report CS-2011-2, E105 - Institut für Stochastik und Wirtschaftsmathematik; Technische Universität Wien, 2011.
- [247] Andreas Alfons, Matthias Templ, and Peter Filzmoser. Simulation of eu-silc population data using simpopulation. Technical Report Europ. Commission, FP7-SSH-2007-21732, WP10-D10.3-3, E105 - Institut für Stochastik und Wirtschaftsmathematik; Technische Universität Wien, 2011.
- [248] Andreas Alfons, Matthias Templ, and Peter Filzmoser. Robust semiparametric estimation of economic indicators from survey samples. Technical Report CS-2011-5, E105 - Institut für Stochastik und Wirtschaftsmathematik; Technische Universität Wien, 2011.
- [249] Andreas Alfons, Matthias Templ, and Peter Filzmoser. Applications of statistical simulation. Technical Report Europ. Commission, FP7-SSH-2007-21732, WP10-D10.3-2, E105 - Institut für Stochastik und Wirtschaftsmathematik; Technische Universität Wien, 2011.
- [250] Andreas Alfons and Matthias Templ. Synthetic data generation. Technical Report European Commission, FP7-SSH-2007-21732, WP6-D6.2-2, E105 - Institut für Stochastik und Wirtschaftsmathematik; Technische Universität Wien, 2011.
- [251] Andreas Alfons and Matthias Templ. Some issues in mapping: Projections and correlations on indicators in time. Technical Report European Commission, FP7-SSH-2007-21732, WP8-D8.2-4, E105 - Institut für Stochastik und Wirtschaftsmathematik; Technische Universität Wien, 2011.
- [252] Andreas Alfons and Matthias Templ. Missing data models for simulation. Technical Report European Commission, FP7-SSH-2007-21732, WP7-D7.1-5, E105 - Institut für Stochastik und Wirtschaftsmathematik; Technische Universität Wien, 2011.
- [253] Andreas Alfons and Matthias Templ. The aat-silc data set. Technical Report European Commission, FP7-SSH-2007-21732, WP6-D6.2-3, E105 - Institut für Stochastik und Wirtschaftsmathematik; Technische Universität Wien, 2011.

- [254] Andreas Alfons, Peter Filzmoser, and Matthias Templ. Summary of the state-of-the-art in visualisation of the european laeken indicators. Technical Report European Commission,FP7-SSH-2007-217322,WP1-D1.1-6, E105 - Institut für Stochastik und Wirtschaftsmathematik; Technische Universität Wien, 2011.
- [255] Andreas Alfons, Peter Filzmoser, Angelika Meraner, and Matthias Templ. Robust distribution fitting. Technical Report European Commission,FP7-SSH-2007-217322,WP4-D4.1-3, E105 - Institut für Stochastik und Wirtschaftsmathematik; Technische Universität Wien, 2011.
- [256] Andreas Alfons, Peter Filzmoser, Stefan Kraft, Matthias Templ, Jan-Philipp Kolb, and Ralf Münnich. Design-based simulations. Technical Report European Commission,FP7-SSH-2007-217322,WP6-D6.1-2, E105 - Institut für Stochastik und Wirtschaftsmathematik; Technische Universität Wien, 2011.
- [257] Andreas Alfons, Peter Filzmoser, Stefan Kraft, and Matthias Templ. The R package: simFrame. Technical Report European Commission,FP7-SSH-2007-217322,WP6-D6.1-3, E105 - Institut für Stochastik und Wirtschaftsmathematik; Technische Universität Wien, 2011.
- [258] Andreas Alfons, Peter Filzmoser, Beat Hulliger, Jan-Philipp Kolb, Stefan Kraft, Ralf Münnich, and Matthias Templ. Synthetic data generation of silc. Technical Report European Commission,FP7-SSH-2007-217322,WP6-D6.2-3, E105 - Institut für Stochastik und Wirtschaftsmathematik; Technische Universität Wien, 2011.
- [259] Andreas Alfons, Jan Pablo Burgard, Peter Filzmoser, Beat Hulliger, Jan-Philipp Kolb, Stefan Kraft, Ralf Münnich, Tobias Schoch, and Matthias Templ. The ameli simulation study. Technical Report European Commission,FP7-SSH-2007-217322,WP6-D6.1, E105 - Institut für Stochastik und Wirtschaftsmathematik; Technische Universität Wien, 2011.
- [260] Matthias Templ, Alexander Kowarik, and Peter Filzmoser. Em-based stepwise regression imputation using standard and robust methods. Technical Report CS-2010-3, E105 - Institut für Stochastik und Wirtschaftsmathematik; Technische Universität Wien, 2010.
- [261] Matthias Templ and Andreas Alfons. Disclosure risk of synthetic population data with application to eu-silc. Technical Report CS-2010-2, E105 - Institut für Stochastik und Wirtschaftsmathematik; Technische Universität Wien, 2010.
- [262] Andreas Alfons, Matthias Templ, and Peter Filzmoser. Simulation of eu-silc population data: using the R package: simPopulation. Technical Report CS-2010-5, E105 - Institut für Stochastik und Wirtschaftsmathematik; Technische Universität Wien, 2010.
- [263] Andreas Alfons, Stefan Kraft, Matthias Templ, and Peter Filzmoser. Simulation of synthetic population data for household surveys with application to eu-silc. Technical Report CS-2010-1, E105 - Institut für Stochastik und Wirtschaftsmathematik; Technische Universität Wien, 2010.
- [264] Matthias Templ, Andreas Alfons, and Peter Filzmoser. Tools for visualising data and aggregated information. Technical Report CS-2009-3, E105 - Institut für Stochastik und Wirtschaftsmathematik; Technische Universität Wien, 2009.
- [265] Matthias Templ and Andreas Alfons. An application of VIM, the R package for visualization of missing values, to EU-SILC data. Technical Report CS-2009-2, E105 - Institut für Stochastik und Wirtschaftsmathematik; Technische Universität Wien, 2009.

- [266] Beat Hulliger, Tobias Schoch, and Matthias Templ. Summary of the state-of-the-art on robustness methods for european laeken indicators. Technical report, E105 - Institut für Stochastik und Wirtschaftsmathematik; Technische Universität Wien, 2009.
- [267] Peter Filzmoser, Karel Hron, and Matthias Templ. Discriminant analysis for compositional data and robust parameter estimation. Technical Report SM-2009-3, E105 - Institut für Stochastik und Wirtschaftsmathematik; Technische Universität Wien, 2009.
- [268] Andreas Alfons, Matthias Templ, and Peter Filzmoser. simFrame: An object-oriented framework for statistical simulation. Technical Report CS-2009-1, E105 - Institut für Stochastik und Wirtschaftsmathematik; Technische Universität Wien, 2009.
- [269] Andreas Alfons, Stefan Kraft, Peter Filzmoser, and Matthias Templ. Progress report on simulation and synthetic data generation for eu-silc data. Technical Report CS-2009-6, E105 - Institut für Stochastik und Wirtschaftsmathematik; Technische Universität Wien, 2009.
- [270] Andreas Alfons, Peter Filzmoser, and Matthias Templ. Summary of the state-of-the-art in visualization of the european laeken indicators. Technical Report AMELI Project Deliverable 8.1, Part 3, E105 - Institut für Stochastik und Wirtschaftsmathematik; Technische Universität Wien, 2009.
- [271] Andreas Alfons, Peter Filzmoser, Stefan Kraft, and Matthias Templ. Summary of the state-of-the-art in simulation in survey statistics. Technical Report AMELI Project Deliverable 6.1, Part1, E105 - Institut für Stochastik und Wirtschaftsmathematik; Technische Universität Wien, 2009.
- [272] Andreas Alfons, Peter Filzmoser, Stefan Kraft, and Matthias Templ. Progress report wp6-simulation. Technical Report AMELI Project Deliverable 6.1, Part 2, E105 - Institut für Stochastik und Wirtschaftsmathematik; Technische Universität Wien, 2009.
- [273] Andreas Alfons, Peter Filzmoser, Stefan Kraft, and Matthias Templ. Intermediate report on the data generation mechanism and of the design of the simulation study. Technical Report AMELI Project Deliverable 6.1, E105 - Institut für Stochastik und Wirtschaftsmathematik; Technische Universität Wien, 2009.
- [274] Andreas Alfons, Peter Filzmoser, Stefan Kraft, and Matthias Templ. Generation of synthetic eu-silc data and simulation. Technical Report CS-2009-5, E105 - Institut für Stochastik und Wirtschaftsmathematik; Technische Universität Wien, 2009.
- [275] Andreas Alfons, Peter Filzmoser, Beat Hulliger, Bernhard Meindl, Tobias Schoch, and Matthias Templ. State-of-the-art in visualization of indicators in survey statistics. Technical Report CS-2009-4, E105 - Institut für Stochastik und Wirtschaftsmathematik; Technische Universität Wien, 2009.
- [276] Matthias Templ and Peter Filzmoser. Visualization of missing values using the r-package vim. Technical Report CS-2008-1, E105 - Institut für Stochastik und Wirtschaftsmathematik; Technische Universität Wien, 2008.
- [277] Matthias Templ. Making sdc-tools better usable by statistical agencies: Sustainability of the argus software. Technical Report CS-2008-5, E105 - Institut für Stochastik und Wirtschaftsmathematik; Technische Universität Wien, 2008.

- [278] Karel Hron, Matthias Templ, and Peter Filzmoser. Imputation of missing values for compositional data using classical and robust methods. Technical Report SM-2008-4, E105 - Institut für Stochastik und Wirtschaftsmathematik; Technische Universität Wien, 2008.
- [279] Matthias Templ. sdcmicro: A package for statistical disclosure control in r. Technical Report CS-2007-5, E105 - Institut für Stochastik und Wirtschaftsmathematik; Technische Universität Wien, 2007.
- [280] Matthias Templ, Peter Filzmoser, and R. Reimann. Cluster analysis applied to regional geochemical data: Problems and possibilities, cs-2006-5. Technical report, E105 - Institut für Stochastik und Wirtschaftsmathematik; Technische Universität Wien, 2006.

Presentations (Keynotes)

- [281] Matthias Templ. Synthetization of complex data and the swiss data anonymization competence center. Keynote: Synth-Data Day, University of Bern, Switzerland, 27.05.2024, 2024.
- [282] M. Templ. Anonymisierung von Daten im Lichte alternativer Konzepte, 2022. IT-Juristinnentag, Keynote, Zurich.
- [283] Matthias Templ. Recent privacy preserving computation or rather traditional methods for protecting personal data? eingeladen; Vortrag: Informatik 2021, Deutschland; 2021-09-29, 2021.
- [284] Matthias Templ. Functional data analysis in Bayes spaces with an application to spatio-temporal population data. Keynote; presentation: The Use of R in Official Statistics - uRos2020, Wien; 2020-12-03, 2020.
- [285] Matthias Templ. Applied compositional data analysis. Keynote; presentation: iCMS2019 - International Conference on Computing, Mathematics and Statistics, Langkawi, Malaysia; 2019-04-23, 2019.
- [286] Matthias Templ. Applied statistical disclosure control: Methods and software. Keynote; presentation: iCMS 2017, Langkawi, Malaysia; 2017-11-06 – 2017-11-09, 2017.
- [287] Matthias Templ. Data visualization using R. Keynote; presentation: eRum 2016, Poznan, Polen; 2016-10-12, 2016.
- [288] Matthias Templ. Essen in den Schlagzeilen: Sound science vs. sounds like science. Keynote; presentation: f.eh im Dialog Ernährungsstudien: Kritik zwischen den Zeilen, Mediatower, Wien; 2015-06-11, 2015.
- [289] Matthias Templ, Alexander Kowarik, and Bernhard Meindl. Development and current practice in using R at Statistics Austria. invited; presentation: New Challenges for Statistical Software - The Use of R in Official Statistics, Bucharest, Romania; 2014-03-27, 2014.
- [290] Peter Filzmoser and Matthias Templ. Data analysis with robust statistical methods. In *International Workshop on Exploratory Data Analysis and Visualization*, page 17, 2010.
- [291] Matthias Templ. Detection of outliers and their influence. Keynote; presentation: Data validation Day, Luxemburg; 2009-11-18, 2009.

Presentations (Invited)

- [292] Matthias Templ. Disclosure risk of anonymized and synthetic data, February 2025. Personalized, data-driven prediction and assessment of Infection related outcomes in Swiss ICUs. IICU Scientific Board Meeting, Zurich.
- [293] Matthias Templ. Anonymization with statistical disclosure control methods using sdcMicro. Workshop held at the Data Protection Week, University of Zurich, 23.01.2025. Available at <https://zenodo.org/records/14764059>, January 2025.
- [294] Matthias Templ. Visualization and imputation of missing values. presentation: DaSSWeb, University of Lisboa, Portugal, 26.4.2024, 2024.
- [295] Matthias Templ. Swiss data anonymization competence center. Statistics & Methodology platform Meeting of the Swiss Clinical Trial Organisation, Invited, 05.05.2024, May 2024.
- [296] Matthias Templ. Introduction to data anonymization. Workshop held on Research Data Management, University of Basel, Switzerland, 24.4.2024, 2024.
- [297] Matthias Templ. Disclosure risk of anonymized and synthetic data. Workshop held at the Clinical Trial Center Zurich, November 2024.
- [298] Matthias Templ. Anonymization of tabular data and swissanon. Invited talk at "Managing Sensitive Personal Data in Research, University of Zurich, 04.11.2024", November 2024.
- [299] Matthias Templ. Anonymization of data with sdcMicro. Invited presentation: Clarin, University of Zurich, Switzerland, 22.3.2024. Available at <https://zenodo.org/records/14191159>, 2024.
- [300] M. Templ. A taxonomy for data anonymization and privacy computing approaches, 2022. ZüKoSt Zürcher Kolloquium über Statistik, ETH Zurich.
- [301] Matthias Templ. Synthetic data. eingeladen; Vortrag: Privacy Technologies for Data Collaboration Synthetic Data, Winterthur; 2021-09-07, 2021.
- [302] Matthias Templ. Population health data anonymisation workshop. invited; presentation: Population Health Data Anonymisation Workshop, Malawi; 2021-04-29, 2021.
- [303] Matthias Templ. Anonymization of personal data with applications in R. eingeladen; Vortrag: Zurich R Courses, Zurich, Switzerland; 2021-11-12, 2021.
- [304] Matthias Templ. Anonymisation of data by synthesising data. invited; presentation: CENTRE FOR STATISTICAL METHODOLOGY series event at the London School for Hygiene and Tropical Medicine, London; 2021-05-26, 2021.
- [305] Matthias Templ. Exploratives lernen mittels fallen stellen, fehler provozieren und feedback geben. invited; presentation: Tag der Lehre, Wädenswil, Schweiz; 2020-01-08, 2020.
- [306] Matthias Templ. Data anonymization. invited; presentation: Gästekolloquium Psychologisches Institut Universität Zürich, Zürich; 2020-11-10, 2020.

- [307] Matthias Templ. Anonymisierung von personenbezogenen Daten für Data Scientists. invited; presentation: Workshop Anonymisierung von personenbezogenen Daten für Data Scientists, Bern; 2020-10-22, 2020.
- [308] Matthias Templ. Workshop on visualization of tidy data. invited; presentation: iCMS2019 - International Conference on Computing, Mathematics and Statistics, Langkawi, Malaysia; 2019-04-23 – 2019-04-24, 2019.
- [309] Matthias Templ. Kompositionsdaten. mit einer anwendung auf dichtefunktionen. presentation: Berufungsvortrag, Bamberg, Deutschland; 2019-06-04, 2019.
- [310] Matthias Templ. Feedback-basiertes, computer-gestuetztes und interaktives unterrichten in der klasse. invited; presentation: Retraite IDP, Winterthur, Switzerland; 2019-06-11, 2019.
- [311] Matthias Templ. Data visualization with r. invited; presentation: 2nd School on Data Analysis and Programming with R, Bolzano, Italy; 2019-08-27, 2019.
- [312] Matthias Templ. Applied compositional data analysis. invited; presentation: SEAMS-UGM 2019, Yogyakarta; 2019-08-01, 2019.
- [313] Matthias Templ. Anonymisierung aus wissenschaftlicher sicht. invited; presentation: Fachtagung Anonymisierung des Vereins fuer Datenschutz, Bern, Schweiz; 2019-08-29, 2019.
- [314] Matthias Templ. Ein feedback-tool zum computergestützten lernen. invited; presentation: IAS-Lehrkolloquium, Waedenswil, Schweiz; 2018-04-11, 2018.
- [315] Matthias Templ. R and data manipulation: Application with big data. invited; presentation: International Conference on Computing, Mathematics and Statistics - Workshop on R and Data Manipulation: Application with Big Data, Langkawi, Malaysia; 2017-11-04 – 2017-11-05, 2017.
- [316] Matthias Templ. The grammar of graphics in r: ggplot2. invited; presentation: R Summer School, Vorau/ Österreich; 2017-09-27, 2017.
- [317] Matthias Templ. Compositional data analysis with an application to our favourite drinks. invited; presentation: Seminar für Statistik, Neuchatel, Switzerland; 2017-10-19, 2017.
- [318] Matthias Templ. Advanced graphics in r. invited; presentation: EURAC Workshop on Advanced R: ggplot2, Bozen; 2017-05-11 – 2017-05-12, 2017.
- [319] Matthias Templ. Simulation of complex synthetic data with the R package simPop. invited; presentation: eRum 2016, Poznan, Polen; 2016-10-12 – 2016-10-14, 2016.
- [320] Matthias Templ. Robust compositional data analysis. invited; presentation: GEOSTAT 2016, Albacete; 2016-09-19 – 2016-09-24, 2016.
- [321] Matthias Templ. Marry me! offizielle statistik & #8734; forschung mittels synthetischer daten. (& methoden der synthetischen datengenerierung). invited; presentation: Mittwoch-Seminar, Statistik Austria, Wien; 2016-03-16, 2016.
- [322] Matthias Templ. Tmf workshop on anonymization tools and their practical relevance. invited; presentation: TMF Workshop on Anonymization tools and their practical relevance, Berlin; 2015-03-19, 2015.

- [323] Matthias Templ. Statistics in practice. why most food and nutrition studies are for the dustbin. Keynote; presentation: 'Teorie a praxe statistického zpracování dat', Pusté Zibřidovice, Czech; 2015-11-26 – 2015-11-28, 2015.
- [324] Matthias Templ. Statistical tables for the employer skills survey of nigeria. invited; presentation: Training Programme for the staff of the Industrial Training Fund of Nigeria on Industrial Skills Gaps Study, Abuja, Nigeria; 2015-10-13 – 2015-10-16, 2015.
- [325] Matthias Templ. The grammar of graphics in r: ggplot2. invited; presentation: School on Data analysis and programming with R, Bozen; 2015-11-18 – 2015-11-20, 2015.
- [326] Matthias Templ, Karel Hron, and Peter Filzmoser. The problem of missing values and rounded zeros in compositional data. invited; presentation: Joint Statistical Meeting, Boston, MA, USA; 2014-08-03, 2014.
- [327] Matthias Templ. Workshop: R in the statistical office. invited; presentation: New Challenges for Statistical Software - The Use of R in Official Statistics, Bucharest, Romania; 2014-03-24, 2014.
- [328] Matthias Templ. Statistik in der praxis. invited; presentation: Hackers Hack Meetup, Museumsquartier, Quartier 21, Halle D; 2014-02-26, 2014.
- [329] Matthias Templ. Software for compositional data analysis. invited; presentation: GeoMap 2014, Olomouc; 2014-06-17, 2014.
- [330] Matthias Templ, Alexander Kowarik, and Francois Fonteneau. The sdcmicro and sdcmicrogui package and automated computations. presentation: Presentation of microdata anonymization open source software and discussion on possible use within and outside the Bank, World Bank, Washington D.C., USA; 2013-05-30, 2013.
- [331] Matthias Templ, Alexander Kowarik, and Florian Endel. R for interactive teaching. invited; presentation: R Summer School, Vorau/ Österreich; 2013-09-25 – 2013-09-27, 2013.
- [332] Matthias Templ and Alexander Kowarik. R crash course. invited; presentation: R Summer School, Vorau/ Österreich; 2013-09-23 – 2013-09-27, 2013.
- [333] Matthias Templ. Providing data with high utility and no disclosure risk for the public and researchers: an evaluation by advanced statistical disclosure risk methods. invited; presentation: CDAM 2013 10th International Conference Computer Data Analysis & Modeling 2013 Theoretical & Applied Stochastics, Minsk, Weißrussland, Belarus; 2013-09-13 – 2013-09-14, 2013.
- [334] Matthias Templ. Log-ratio analyse von bieren, whiskies und kaffee's und eine praktische anwendung an doktoratsstudenten. presentation: Seminar Statistik, Technische Universität Graz; 2013-10-16, 2013.
- [335] Matthias Templ. Estimation and visualisation of indicators: Applications from social statistics. invited; presentation: Workshop 'Teorie a praxe statistického zpracování dat', Velke Losiny, CZ; 2013-11-21, 2013.
- [336] Alexander Kowarik and Matthias Templ. Visualisation with ggplot2. invited; presentation: R Summer School, Vorau/ Österreich; 2013-09-24 – 2013-09-27, 2013.

- [337] A Kalidova, Karel Hron, Matthias Templ, and Peter Filzmoser. Replacement of missing values and rounded zeros in high-dimensional compositional data with application to metabolomics. presentation: ODAM 2013 - Olomoucian Days of Applied Mathematics, Olomouc; 2013-06-12 – 2013-06-14, 2013.
- [338] Andreas Alfons and Matthias Templ. Sweave, knitr, rstudio, r markup. invited; presentation: R Summer School, Vorau/ Österreich; 2013-09-24 – 2013-09-27, 2013.
- [339] Matthias Templ and Andreas Alfons. Survey methodology and simulation in r. possibilities and new methods. presentation: Föschungsseminar des Instituts für Angewandte Statistik der Johannes Kepler Universität Linz, Linz; 2010-11-18, 2010.
- [340] Matthias Templ and Andreas Alfons. Statistical microsimulations with complex surveys and application to social inclusion indicators. presentation: Diskussionsforum Junge Statistik, Wien; 2010-12-02, 2010.
- [341] Matthias Templ. Discussion points for the panel session on future direction in data analysis and visualization. invited; presentation: Workshop on Exploratory Data Analysis and Visualization, Wien; 2010-05-27 – 2010-05-28, 2010.
- [342] Matthias Templ. Visualisierung von fehlenden werten. invited; presentation: 18. Wissenschaftliches Kolloquium über Informationsvisualisierung, Wiesbaden; 2009-11-20, 2009.
- [343] Andreas Alfons, Matthias Templ, and Peter Filzmoser. Complex survey data sets: Visualization of missing values in r. invited; presentation: Young European Statisticians Workshop, Eindhoven; 2008-10-07, 2008.
- [344] Matthias Templ. Statistical disclosure control (sdc) in r: Generation of anonymized data for researchers and the relation to robust statistics. invited; presentation: International Workshop on Robust Statistics and R, Banff; 2007-11-01, 2007.
- [345] Heinrich Fritz, Peter Filzmoser, and Matthias Templ. Robust estimation of missing values. invited; presentation: International Workshop on Robust Statistics and R, Banff; 2007-11-01, 2007.

Presentations (Others)

- [346] Oscar Thees, Jiří Novák, and Matthias Templ. Evaluation of synthetic data generators on complex tabular data. Presentation at Privacy in Statistical Databases, Antibes, France, 2024.
- [347] P. Jaskova, Matthias Templ, D. Lachman, Karel Hron, B. Magali, and J. Palarea-Albaladejo. Pairwise logratio analysis in high-dimensional context. presentation: CoDaWork 2024, Girona, Spain, 2024.
- [348] C. Sluka. Anonymisation of clinical data for the swiss stroke registry. Master's thesis, ZHAW, Winterthur, Schweiz, 2023. supervisor: M. Templ.
- [349] Matthias Templ. Ist-stand des austrian journal of statistics. presentation: Vorstandssitzung Oesterreichische Statistische Gesellschaft, Wien; 2021-02-18, 2021.

- [350] Matthias Templ. Interaktive grafiken. eingeladen; Vortrag: Schweizer Statistiktage 2021, Lugano, Schweiz; 2021-09-06, 2021.
- [351] Matthias Templ. Can we ignore the compositional nature of compositional data by using deep learning aproaches? eingeladen; Vortrag: SIS 2021, Sardinia; 2021-02-28, 2021.
- [352] Matthias Templ. Anonymizing data using sdc. eingeladen; Vortrag: BernR User Group, Bern, Schweiz; 2021-11-11, 2021.
- [353] Matthias Templ. Anonymisierung smart metering daten konzept langzeitarchivierung. presentation: Steering Board Meeting, Energiewirtschaft Schweiz, Zurich; 2021-01-21, 2021.
- [354] Thoralf Mildenerger, Matthias Templ, Andreas Papritz, and Flavio Schönholzer. Ad-hoc-stichprobe über wohnungsmieten in der stadt zürich - ein design-basierter ansatz zur verbindung von zufalls- und ad-hoc-stichproben. Vortrag: Schweizer Statistiktage 2021, Lugano, Schweiz; 2021-09-06, 2021.
- [355] Johannes Gussenbauer, Matthias Templ, Siro Fritzmann, and Alexander Kowarik. Synthetic data with xgboost and advanced calibration. Vortrag: The use of R in official statistics. uRos 2021, Bukarest; 2021-11-25, 2021.
- [356] J. Gussenbauer, M. Templ, S. Fritzmann, and A. Kowarik. Synthetic data with xgboost and advanced calibration. In *uros 2021*, 2021.
- [357] Peter Filzmoser, Sugnet Lubbe, and Matthias Templ. Strategies to replace high proportions of zeros in compositional data. invited; presentation: Online Conference - 1st Conference on Information Technology and Data Science, Debrecen; 2020-11-07, 2020.
- [358] Matthias Templ and Murat Sariyar. Workshop zur Anonymisierung von personenbezogenen Einzeldaten. presentation: Workshop zur Anonymisierung von personenbezogenen Einzeldaten, Biel, Schweiz; 2019-09-30, 2019.
- [359] Matthias Templ. Imputation von fehlenden werten mit deep learning methoden. presentation: IDP Kolloquium, Winterthur; 2019-03-03, 2019.
- [360] Matthias Templ. Feedback-basiertes, computer-gestuetztes und interaktives unterrichten. presentation: Abschlussveranstaltung Digital Learning, Winterthur, Switzerland; 2019-01-24, 2019.
- [361] Matthias Templ. Computergestuetztes feedback-basiertes unterrichten. invited; presentation: Statistikertreffen ZHAW, Winterthur, Switzerland; 2019-06-18, 2019.
- [362] Matthias Templ and Rene Locher. Data analysis and statistics at the idp. presentation: ZHAW / UiTM Kedah Meeting, Winterthur, Switzerland; 2018-02-05, 2018.
- [363] Nikos Tzavidis, Timo Schmid, Natalia Rojas, Soeren Pannier, Ann-Kristin Kreuzmann, and Matthias Templ. Model-based approaches to poverty mapping: A critical review. presentation: Final Conference - InGRID Project. Better science infrastructure for evidence-based policies on inclusive growth in Europe, Brussels; 2017-01-17, 2017.

- [364] Nikos Tzavidis, Soeren Pannier, Natalia Rojas, Timo Schmid, and Matthias Templ. emdi: An R package for estimating and mapping disaggregated indicators. presentation: NCRM Annual Centre Meeting, Southampton; 2017-01-17, 2017.
- [365] Matthias Templ and Alexander Kowarik. Simulation von public-use, realitätsnahen, synthetischen populationen. presentation: SurveyConf 2017, Gumpoldskirchen; 2017-05-24 – 2017-05-27, 2017.
- [366] Matthias Templ and Alexander Kowarik. Simulation of public-use complex survey and population data. presentation: ISI 2017, Marrakech; 2017-07-16 – 2017-07-20, 2017.
- [367] Matthias Templ, Karel Hron, and Peter Filzmoser. Exploring outliers in compositional data with structural zeros. presentation: CoDaWork 2017, Abbadia San Salvatore; 2017-06-06 – 2017-06-09, 2017.
- [368] Matthias Templ. Simulation von synthetischen populationen fuer agentenbasierte microsimulationen und open-data access. invited; presentation: IDP Kolloquium, Winterthur; 2017-03-09, 2017.
- [369] Matthias Templ. Schulung zur Anonymisierung von personenbezogenen Datenbeständen. presentation: Schulung zur Anonymisierung von personenbezogenen Datenbeständen, Biel, Schweiz; 2017-06-22, 2017.
- [370] Matthias Templ. Relevante begriffe in der statistik. presentation: SIA-Erhebungen, Zurich; 2017-08-21, 2017.
- [371] Matthias Templ. Computergestütztes statistiktraining mit erzwungenem instant-feedback. presentation: Treffen der an Statistik interessierten Personen der ZHAW, Winterthur; 2017-06-20, 2017.
- [372] Matthias Templ. Computergestueztes statistiktraining mit erzwungenem instant-feedback das lehr-und feedbacksystem shinyfeedback. invited; presentation: Treffpunkt Statistik, Winterthur, Switzerland; 2017-06-20, 2017.
- [373] Matthias Templ. Compositional data analysis, probability density functions and functional data analysis. with an application to spatio-temporal population pyramids. presentation: IDP Kolloquium, Winterthur; 2017-03-01, 2017.
- [374] Alexander Kowarik and Matthias Templ. Visualisation and imputation of missing values. presentation: ISI 2017, Marrakech; 2017-07-16 – 2017-07-20, 2017.
- [375] Roman Etschmann, Matthias Templ, and Christoph Heitz. Entwicklung eines stationsungebundenen e-bike-sharing-services mit nutzer-coproduction. presentation: IDP Kolloquium, Winterthur; 2017-04-10, 2017.
- [376] Matthias Templ. Die gesetze der grossen zahlen. presentation: ZHAW Seminar fuer Statistik, Winterthur; 2016-05-10, 2016.
- [377] Matthias Templ. Data visualization using R. Keynote; presentation: eRum 2016, Poznan, Polen; 2016-10-12, 2016.

- [378] Matthias Templ. Analyse relativer information. presentation: ZHAW Seminar fuer Statistik, Winterthur; 2016-05-10, 2016.
- [379] Karel Hron, Matthias Templ, and Peter Filzmoser. Exploring outliers in compositional data with structural zeros. presentation: International Conference of the ERCIM WG on Computational and Methodological Statistics, Seville, Spain; 2016-12-09 – 2016-12-11, 2016.
- [380] Matthias Templ and Valentin Todorov. Tutorial: R in the statistical office. invited; presentation: NTTS 2015, Brussels, Belgium; 2015-03-13, 2015.
- [381] Matthias Templ, Karel Hron, Alessandra Menafoglio, Klara Hruskova, and Peter Filzmoser. Simplicial principal component analysis for density functions. presentation: CoDaWork 2015, L'Escala; 2015-06-01 – 2015-06-05, 2015.
- [382] Matthias Templ, Johannes Gussenbauer, Peter Filzmoser, and Olivier Dupriez. Outlier detection in complex survey data including semi-continuous components and missing values. invited; presentation: ERCIM 2015, London; 2015-12-11 – 2015-12-14, 2015.
- [383] Matthias Templ. Visualization of spatial information using ggmap. presentation: Seminar on Statistics, Palacky University of Olomouc; 2015-04-13, 2015.
- [384] Matthias Templ. Traditional methods for statistical disclosure control. presentation: Working Group Meeting according to the Special Grant Agreement on Public-Use Files of the European Labour Force Survey and the Statistics on Income and Living Conditions, Luxembourg; 2015-03-03, 2015.
- [385] Matthias Templ. Synthetic data: Modelling and generating complex close-to-reality data for public use. presentation: Olomoucian Days of Applied Mathematics, Olomouc; 2015-05-21, 2015.
- [386] Matthias Templ. R verstehen lernen. presentation: R Kurs, TU Wien; 2015-04-08 – 2015-04-09, 2015.
- [387] Matthias Templ. R packaging and new development features for building R packages. presentation: Seminar on Statistics, Palacky University of Olomouc; 2015-02-25, 2015.
- [388] Matthias Templ. Overview existing methodology pufs in general and silc/lfs specific. presentation: First Full Project meeting of the SGA on PUFs of EU-SILC and EU-LFS, Luxembourg; 2015-03-02, 2015.
- [389] Matthias Templ. Outlier detection. invited; presentation: OMAN Study Visit at UNIDO, UNIDO; 2015-08-04, 2015.
- [390] Matthias Templ. Most important variables and indicators for eu-silc and lfs related to users' needs. presentation: Working Group Meeting according to the Special Grant Agreement on Public-Use Files of the European Labour Force Survey and the Statistics on Income and Living Conditions, Luxembourg; 2015-03-03, 2015.
- [391] Matthias Templ. Methods to create synthetic public-use files. presentation: Working Group Meeting according to the Special Grant Agreement on Public-Use Files of the European Labour

- Force Survey and the Statistics on Income and Living Conditions, Luxembourg; 2015-03-03, 2015.
- [392] Matthias Templ. Forschungsschwerpunkte in der offiziellen statistik und stichprobentheorie. presentation: SurveyConf 2015, Berlin; 2015-10-01 – 2015-10-04, 2015.
 - [393] Matthias Templ. Designing plots with ggplot2. presentation: Seminar on Statistics, Palacky University of Olomouc; 2015-04-13, 2015.
 - [394] Matthias Templ. Analysis of incomplete data and imputation of missing values: the R package VIM. presentation: missData 2015, Rennes, France; 2015-06-18, 2015.
 - [395] Barbara Szabo, Matthias Templ, Peter Filzmoser, Annamária Lehoczky, Rita Pongrácz, et al. Biogeographical regions and their flowering phenological patterns across europe. presentation: Phenology 2015, Kusadasi; 2015-10-05 – 2015-10-08, 2015.
 - [396] Bernhard Meindl and Matthias Templ. Producing synthetic and confidential data using r-package simpop. presentation: 2nd European Data Access Forum, Luxembourg; 2015-03-24 – 2015-03-25, 2015.
 - [397] Johannes Gussenbauer, Peter Filzmoser, Matthias Templ, and Olivier Dupriez. Robust statistical methods for outlier detection with application to household expenditure data. presentation: Statistiktage 2015, Wien; 2015-10-26 – 2015-10-28, 2015.
 - [398] Alzbetka Gardlo, Karel Hron, Matthias Templ, and Peter Filzmoser. Imputation of rounded zeros for data from metabolomics. presentation: CoDaWork 2015, L'Escala; 2015-06-01 – 2015-06-05, 2015.
 - [399] Raimon Tolosana-Delgado, K.G. van den Boogaart, Peter Filzmoser, Karel Hron, and Matthias Templ. Compositional regression: an overview. presentation: IAMG2014 - 16th Annual Conference of the International Association for Mathematical Geosciences, New Delhi; 2014-10-17 – 2014-10-20, 2014.
 - [400] Matthias Templ and Valentin Todorov. R in the statistical office. invited; presentation: European Conference on Quality in Official Statistics (Q2014), Vienna; 2014-05-02, 2014.
 - [401] Matthias Templ and Bernhard Meindl. Methods and tools for the generation of synthetic populations. presentation: Privacy in Statistical Databases 2014, Ibiza, Spain; 2014-09-17 – 2014-09-19, 2014.
 - [402] Matthias Templ, Alexander Kowarik, and Bernhard Meindl. R verstehen lernen. presentation: R Kurs Wirtschaftskammer Oesterreich, WKO Wien; 2014-05-12 – 2014-05-13, 2014.
 - [403] Matthias Templ and Alexander Kowarik. Visualization and imputation of missing values. presentation: Erste deutschsprachige Survey-Statistik-Tagung, Zeltingen-Rachtig; 2014-04-07 – 2014-04-09, 2014.
 - [404] Matthias Templ and Alexander Kowarik. Imputation of missing values with the R package VIM. presentation: useR! 2014, Los Angeles, USA; 2014-07-02, 2014.

- [405] Matthias Templ and Alexander Kowarik. Forschungsbereiche in der offiziellen statistik und stichprobentheorie. invited; presentation: Erste deutschsprachige Survey-Statistik-Tagung, Zeltingen-Rachtig; 2014-04-07 – 2014-04-09, 2014.
- [406] Matthias Templ, Andreas Alfons, and Peter Filzmoser. Robust estimation of income and social indicators with tail modelling. invited; presentation: Seminar in Applied Mathematics, Palacky University Olomouc; 2014-01-27, 2014.
- [407] Matthias Templ. Statistical computing at tu wien. presentation: TEMPUS international project meeting, Tu Wien; 2014-05-13, 2014.
- [408] Matthias Templ. Some advanced and useful development tools in r. invited; presentation: Seminar on Statistics, Palacky University of Olomouc; 2014-10-20, 2014.
- [409] Matthias Templ. Simulation of synthetic populations. invited; presentation: Seminar in Applied Mathematics (2), Palacky University Olomouc, Czech Republik; 2014-05-19, 2014.
- [410] Matthias Templ. R development. invited; presentation: R Meetup 2014/I, TU Wien; 2014-02-11, 2014.
- [411] Barbara Szabo, Attila Lengyel, Matthias Templ, and Bálint Czúcz. Improving classification of plant assemblages with environmental constraints: a simulation study. Posterpräsentation: European Vegetation Survey, 23rd Workshop, Lubijana; 2014-05-08 – 2014-05-12, 2014.
- [412] Barbara Szabo, Annamária Lehoczky, Peter Filzmoser, Matthias Templ, Ferenc Szentkirályi, Rita Pongrácz, Thomas Ortner, Mehmet Can Mert, Bálint Czúcz, et al. From south to north: flowering phenological responses at different geographical latitudes in 12 european countries. Posterpräsentation: European Geosciences Union General Assembly 2014, Wien; 2014-04-27 – 2014-05-02, 2014.
- [413] Bernhard Meindl, Alexander Kowarik, and Matthias Templ. sparktable: Generating graphical tables for websites and documents with r. presentation: New Challenges for Statistical Software - The Use of R in Official Statistics, Bucharest, Romania; 2014-03-23, 2014.
- [414] Alexander Kowarik and Matthias Templ. New features of vim. presentation: UNECE worksession on statistical data editing, Paris; 2014-04-28, 2014.
- [415] Alexander Kowarik, Bernhard Meindl, and Matthias Templ. sparktable - generating graphical tables for websites and documents with r. presentation: European Conference on Quality in Official Statistics (Q2014), Vienna; 2014-05-04, 2014.
- [416] Matthias Templ, Alexander Kowarik, and Bernhard Meindl. Anonymization of microdata. invited; presentation: Österreichische Statistiktage 2013, Statistik Austria, Wien; 2013-10-23 – 2013-10-25, 2013.
- [417] Matthias Templ. Statistical disclosure control for microdata. presentation: StatGIS Seminar, Tschechien, Olomouc; 2013-10-01, 2013.
- [418] Matthias Templ. S4 class sdcmicro. presentation: Expert Group SDC, Luxembourg; 2013-10-08, 2013.

- [419] Matthias Templ. Outlier detection in compositional data with structural zeros. presentation: ODAM 2013 - Olomoucian Days of Applied Mathematics, Olomouc; 2013-06-13 – 2013-06-14, 2013.
- [420] Matthias Templ. Basic concepts of imputation. presentation: Statistics Seminar at the Palacký University, Olomouc, Czech Republic; 2013-01-14, 2013.
- [421] Matthias Templ. Anonymisation of confidential data; methods and software. invited; presentation: Österreichische Statistiktage 2013, Statistik Austria, Wien; 2013-10-23 – 2013-10-25, 2013.
- [422] Matthias Templ. Advanced imputation methods. presentation: Statistics Seminar at the Palacký University, Olomouc, Czech Republic; 2013-03-11, 2013.
- [423] Gianna Monti, Karel Hron, Peter Filzmoser, and Matthias Templ. Covariance-based outlier detection for compositional data with structural zeros: Application to italian survey of household income and wealth data. presentation: SIS 2013 - Advances in Latent Variables - Methods, Models and Applications, Brescia, Italy; 2013-06-21, 2013.
- [424] Alexander Kowarik and Matthias Templ. Make your data confidential with the `sdcmicro` and `sdcmicrogui` packages. presentation: The R User Conference, useR! 2013, University of Castilla-La Mancha, Albacete, Spain; 2013-07-11 – 2013-07-12, 2013.
- [425] Alexander Kowarik, Angelika Meraner, and Matthias Templ. Interactive adjustment and outlier detection of time dependent data in `r` - `x12`, `x12gui`. presentation: Österreichische Statistiktage 2013, Statistik Austria, Wien; 2013-10-25, 2013.
- [426] Matthias Templ and Valentin Todorov. Tutorium: R für statistische organisationen. invited; presentation: Statistische Woche 2012, TU Wien; 2012-09-18 – 2012-09-21, 2012.
- [427] Matthias Templ, Alexander Kowarik, Bernhard Meindl, and Francois Fonteneau. Testing of `ihsn c++` code and inclusion of new methods into `sdcmicro`. presentation: Konferenz: Privacy in Statistical Databases (PSD 2012), Palermo/ Sizilien/ Italien; 2012-09-26 – 2012-09-28, 2012.
- [428] Matthias Templ, Alexander Kowarik, and Beat Hulliger. Visualization of complex statistical information in maps: The checkerplot. invited; presentation: Joint Statistical Meeting, San Diego; 2012-07-28 – 2012-08-02, 2012.
- [429] Matthias Templ and Thomas Burg. Methodische entwicklungen im bereich indikatorschätzung - ausgewählte resultate aus dem projekt ameli. presentation: Seminar, Statistik Austria, Wien; 2012-03-28, 2012.
- [430] Matthias Templ. Von erhebungsdaten zur erkenntnis (habilitationsvorstellung). invited; presentation: Habilitationsvorstellung, TU Wien; 2012-03-20, 2012.
- [431] Matthias Templ. Standard software in sdc for oecd and non-oecd countries: `sdcmicro` (and `sdcmicrogui`). invited; presentation: Expert Working Group on Statistical Disclosure Control, Luxembourg; 2012-12-11, 2012.
- [432] Matthias Templ. New methods to present regional indicators in maps and graphical tables. presentation: Universität Olomouc, Czech Republic; 2012-11-26, 2012.

- [433] C Poirier and Matthias Templ. Introduction to topic v - software & tools for data editing and imputation. presentation: Conference of European Statisticians, Oslo / Norwegen; 2012-09-25, 2012.
- [434] Karel Hron, Peter Filzmoser, Matthias Templ, G van den Boogaart, and Raimon Tolosana-Delgado. A unified approach to classical and robust regression for compositional data. presentation: 5th International Conference of the ERCIM WG on COMPUTING & STATISTICS, Oviedo / Spain; 2012-12-03, 2012.
- [435] Matthias Templ, Alexander Kowarik, Peter Filzmoser, and Andreas Alfons. A computational and methodological framework for visualisation and imputation of missing values: the r-package vim. presentation: Statistische Tage 2011, Graz; 2011-09-09, 2011.
- [436] Matthias Templ, Andreas Alfons, and Peter Filzmoser. Robust semi-parametric estimation of economic and social indicators. invited; presentation: 58th Session of the ISI Conference, Dublin; 2011-08-26, 2011.
- [437] Matthias Templ. Presenting aggregated information in maps and tables: Visualization suggestions from the ameli project. presentation: University of Applied Sciences, Northwestern Switzerland, Olten; 2011-03-15, 2011.
- [438] Matthias Templ. New methods and software for data editing. invited; presentation: UNECE worksession on statistical data editing, Ljubljana; 2011-05-09 – 2011-05-11, 2011.
- [439] Bernhard Meindl and Matthias Templ. A computational framework to protect tabular data - r-package sdctable. invited; presentation: Conference of European Statisticians in Tarragona, Joint UNECE/Eurostat work session on statistical data confidentiality, Spanien; 2011-10-26 – 2011-10-28, 2011.
- [440] Gerlinde Dinges, Alexander Kowarik, Bernhard Meindl, and Matthias Templ. Statistikunterricht interaktiv mit tgui (teaching graphical user interface). invited; presentation: Tag der amtlichen Statistik, Wien; 2011-10-20, 2011.
- [441] Stefan Zechner, Peter Filzmoser, Matthias Templ, and Andreas Alfons. Visualization of indicators in R with application to EU-SILC. invited; presentation: 4th AMELI Meeting, Wien; 2010-02-24, 2010.
- [442] Matthias Templ, Alexander Kowarik, and Peter Filzmoser. Irm: An open-source solution for imputation of complex data using robust methods. presentation: European Conference on Quality in Official Statistics 2010, Helsinki; 2010-05-05, 2010.
- [443] Matthias Templ and Alois Haslinger. Opening of the edavis workshop. invited; presentation: Workshop on Exploratory Data Analysis and Visualization, Wien; 2010-05-27 – 2010-05-28, 2010.
- [444] Matthias Templ, Gerlinde Dinges, Bernhard Meindl, and Alexander Kowarik. New ways of lecturing: Statistics austria's experience and vienna university of technology's interest on a master programme. invited; presentation: University of Southampton, United Kingdom; 2010-02-24, 2010.

- [445] Matthias Templ and Andreas Alfons. A short overview of the simulation facilities included in deliverable 6.1. invited; presentation: AMELI Meeting in Wien, Wien; 2010-02-23, 2010.
- [446] Matthias Templ and Andreas Alfons. Armutsschätzung auf das intervall gebracht. innovationen aus wien im zuge des ameli projekts. presentation: Österreichische Statistiktage 2010, Wien; 2010-10-22, 2010.
- [447] Matthias Templ. R für lehrende. invited; presentation: Workshop on Exploratory Data Analysis and Visualization, Wien; 2010-05-27 – 2010-05-28, 2010.
- [448] Matthias Templ. Opening of the international conference on indicators and survey methodology. invited; presentation: Intenational Conference on Indicators and Survey Methodology, Wien; 2010-02-25, 2010.
- [449] Matthias Templ. Closing remarks for the international conference on indicators and survey methodology. invited; presentation: International Conference on Indicators and Survey Methodology, Wien; 2010-02-25, 2010.
- [450] Matthias Templ. Automatische generierung von reports mit latex und r. invited; presentation: Workshop on Exploratory Data Analysis and Visualization, Wien; 2010-05-27 – 2010-05-28, 2010.
- [451] Angelika Meraner, Matthias Templ, and Peter Filzmoser. Outlier detection for semi-continuous variables. invited; presentation: AMELI Meeting in Wien, Wien; 2010-02-24, 2010.
- [452] Andreas Alfons, Matthias Templ, and Peter Filzmoser. Text von andi. invited; presentation: AMELI Meeting in Wien, Wien; 2010-02-23, 2010.
- [453] Andreas Alfons, Matthias Templ, and Peter Filzmoser. Pareto tail modeling for social inclusion indicators. invited; presentation: 6th AMELI Meeting, Trier; 2010-10-04, 2010.
- [454] Andreas Alfons, Matthias Templ, and Peter Filzmoser. Application of the R package simFrame for statistical simulation to EU-SILC. invited; presentation: 4th AMELI Meeting, Wien; 2010-02-24, 2010.
- [455] Matthias Templ and Alexander Kowarik. Robust imputation of data consisting of nominal, ordinal, semi-continuous and continuous variables or compositional data. invited; presentation: 3rd AMELI meeting, Olten; 2009-06-04, 2009.
- [456] Matthias Templ, Josef Holzer, Peter Filzmoser, and Andreas Alfons. Robust methods for the estimation of selected laeken indicators. invited; presentation: 3rd AMELI meeting, Olten; 2009-06-04, 2009.
- [457] Matthias Templ and Andreas Alfons. Die verwendung von mikrodaten in der wissenschaftlichen lehre. invited; presentation: Workshop zur Nutzung von SDS in der wissenschaftlichen Lehre, Wien; 2009-06-17, 2009.
- [458] Matthias Templ. Die verwendung von mikrodaten in der wissenschaftlichen lehre. presentation: Workshop bei der Statistik Austria, Wien; 2009-06-17, 2009.

- [459] Bernhard Meindl, Matthias Templ, and Alois Haslinger. Confidentiality issues at statistics austria. invited; presentation: Meeting on methodological work in Statistics Austria and Statistics New Zealand, Wien; 2009-10-15, 2009.
- [460] Stefan Kraft, Peter Filzmoser, and Matthias Templ. Simulation of a population for the european living and income conditions survey. invited; presentation: 3rd AMELI meeting, Olten; 2009-06-05, 2009.
- [461] Gerlinde Dinges, Alexander Kowarik, Bernhard Meindl, and Matthias Templ. Neue wege der kursgestaltung. invited; presentation: Meeting des Bereiches Register, Klassifikation und Methodik, Statistik Austria, Wien; 2009-10-12, 2009.
- [462] Andreas Alfons, Matthias Templ, and Peter Filzmoser. State-of-the-art and recent developments in visualisation of missing values and indicators. invited; presentation: 3rd AMELI meeting, Olten; 2009-06-05, 2009.
- [463] Andreas Alfons and Matthias Templ. Design-based simulation using the R-package simFrame. invited; presentation: 3rd AMELI meeting, Olten; 2009-06-05, 2009.
- [464] Matthias Templ and Bernhard Meindl. Robustification of microdata masking methods. presentation: Privacy in Statistical Databases, Istanbul; 2008-09-25, 2008.
- [465] Matthias Templ and Bernhard Meindl. Robust statistics meets SDC. presentation: Privacy in Statistical Databases, Istanbul; 2008-09-26, 2008.
- [466] Matthias Templ and Josef Holzer. Several aspects about robustness in the ameli project. presentation: AMELI Meeting, Wiesbaden; 2008-11-06, 2008.
- [467] Matthias Templ, Peter Filzmoser, and Clemens Reimann. Cluster analysis: Data preparation? which algorithm? how many clusters? invited; presentation: Lifestat 2008, München; 2008-03-12, 2008.
- [468] Matthias Templ and Peter Filzmoser. Visualization and the use of r-forge for collaborative research projects. invited; presentation: AMELI kick-off meeting, Neuchatel; 2008-04-17 – 2008-04-18, 2008.
- [469] Matthias Templ, Andreas Alfons, and Peter Filzmoser. Recent developments in visualization : Focus on missing values and maps. presentation: AMELI Meeting, Wiesbaden; 2008-11-07, 2008.
- [470] Matthias Templ. Statistik wozu? einige moderne anwendungen. presentation: Welcome Day 2008, Wien; 2008-10-01, 2008.
- [471] Matthias Templ. Simulation study and the generation of universes. invited; presentation: AMELI kick-off meeting, Neuchatel; 2008-04-17 – 2008-04-18, 2008.
- [472] Stefan Kraft, Matthias Templ, and Peter Filzmoser. Simulation. presentation: AMELI Meeting, Wiesbaden; 2008-11-07, 2008.
- [473] Andreas Alfons, Matthias Templ, and Peter Filzmoser. Visualization software for eu-silc data and laeken indicators. invited; presentation: AMELI Meeting, Wiesbaden; 2008-11-07, 2008.

- [474] Matthias Templ, Peter Filzmoser, and Clemens Reimann. Problems and possibilities of cluster analysis: Application on geochemical data. invited; presentation: ROeS Seminar 2007, Bern; 2007-09-10, 2007.
- [475] Matthias Templ and Peter Filzmoser. Visualisation of missing values and robust imputation in environmental surveys. invited; presentation: TIES 2007, 18th annual meeting of the International Environmetrics Society, Mikulov; 2007-08-16, 2007.
- [476] Matthias Templ and Peter Filzmoser. Stability of cluster analysis. invited; presentation: The R User Conference 2006, Wien; 2006-06-16, 2006.
- [477] Matthias Templ and Gerlinde Dinges. Motivation zur statistik: Computergestütztes lernen in der statistik austria. presentation: Austrian Statistical Society, Vienna; 2006-10-05, 2006.
- [478] Matthias Templ. Statistical data disclosure: Methods and software development. invited; presentation: Charles University, Prag; 2006-10-04, 2006.
- [479] Matthias Templ. Using r for data disclosure. presentation: Statistische Tage, Wien; 2004-10-01, 2004.
- [480] Matthias Templ. Cluster analysis with application to geochemical data. presentation: RMED'03: Workshop on Robust Modeling of Environmental Data, Vorau; 2003-07-09, 2003.
- [481] Matthias Templ. Cluster analysis und ihre anwendung auf geochemische daten. presentation: Österreichische Statistiktage 2003, Wien; 2003-10-30, 2003.
- [482] P. Jaskova, M. Templ, and K. Hron. Forward selection with orthogonal pairwise logratios in regression with compositional covariates. CoDaWork2022, Toulouse, France.
- [483] J. Gussenbauer, A. Kowarik, S. Fritzmann, and M. Templ. Synthetic data generation with simPop: New features XGBoost and advanced calibration. useR 2022.
- [484] C. Gozzi, M. Templ, and A. Buccianti. The langelier-ludwig square diagram through a new compositional lens. The Geoscience paradigm: resources, risk and future perspectives, Potenza, Italy.

Miscellaneous

- [0] Matthias Templ. Anonymization course for the high-level management. invited; presentation: Workshop Anonymization of personal data for the high-level management, Bern; 2020-10-18, 2020.
- [0] Matthias Templ. Workshop on data visualisation in r. invited; presentation: UGM 2019, Yogyakarta, Indonesia; 2019-07-27, 2019.
- [0] Matthias Templ. Game of lazybones computer-assisted exercises with automatised feedback and evaluation. invited; presentation: Statistics at Universities of Applied Sciences (UAS), Olten, Schweiz; 2018-01-26, 2018.
- [0] Matthias Templ. Compositional data analysis of our favourite drinks. invited; presentation: eRum 2018, Budapest; 2018-05-13 – 2018-05-15, 2018.

- [0] Matthias Templ. Creating public-use synthetic data from complex surveys. invited; presentation: Swiss Statistics Seminar, Bern; 2017-05-05, 2017.
- [0] Matthias Templ. Data visualization. invited; presentation: School on Data analysis and programming with R, Bozen; 2015-11-18 – 2015-11-20, 2015.
- [0] Matthias Templ. Fantastic four: sdcmicro, sdcmicrogui, sdctable, synthpop. invited; presentation: Expert Group Meeting on Statistical Disclosure Control, Luxembourg; 2014-06-24, 2014.
- [0] Matthias Templ and Alexander Kowarik. Visualisation with r - graphics, lattice, colors. presentation: R Summer School, Vorau/ Österreich; 2013-09-24 – 2013-09-27, 2013.
- [0] Matthias Templ. Log-ratio analysis of the chemical composition of (our) favoured drinks. invited; presentation: Workshop 'Teorie a praxe statistického zpracování dat', Dolni Morava, Czech; 2013-11-14 – 2013-11-16, 2013.
- [0] Matthias Templ, Bernhard Meindl, and Alexander Kowarik. Statistical disclosure control using sdcmicro and sdcmicrogui. invited; presentation: OECD, Workshop on SDC, Paris; 2012-08-12, 2012.
- [0] Matthias Templ. Teaching statistics with r : Beyond e-learning systems. invited; presentation: Workshop "R" in Teaching and Empirical Research, Wien; 2009-01-16, 2009.